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MA Thesis New Media and Digital Culture

Frameworks of Legitimation in the NFT Space: Decentralizing the Diffusion of Art through Platforms

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Abstract: The NFT market, founded on the developments of blockchain technologies that enable decentralization and disintermediation, has imposed new rules, genres and styles on media art, consolidating the genre of crypto art. Platforms built on different blockchains have taken the lead managing the new market, whose dynamics are determined in part by the technologies that support them, and in part by the design choices exercised by their creators. If the contemporary art system, for its current market, has shaped legitimation through the interaction of dominant voices (such as critics, art-dealers, galleries and entrepreneurs) who play a central role in its dissemination as central actors of its supply chain, this new market seems to be challenging these traditional frameworks. The thesis studies two platforms that during the NFT boom in 2021 attracted most NFT enthusiasts: Opensea and Hic et Nunc. Of the two platforms, the study analyzes through a social aesthetics approach: the characteristics of the works that are created, promoted, and exchanged; the ways in which artists are presented; and the governance that determines the mechanisms and processes of mediation of the two platforms.

The thesis investigates the frameworks of legitimation proposed by the two substantially diverse models of Opensea and Hic et Nunc, platforms in which the artist is also in turn transforming himself: in most cases he is anonymous; relies on the generativity of technology to conclude or multiply his work; accepts community evaluations and algorithmic calculations of navigational models proposed by the platforms which inevitably reflect on the social contexts that flourish within this ecosystem. The thesis will demonstrate how among these two different social contexts, Opensea, relying on an economic and profit oriented conception of art as commodity, appears to propose legitimation frameworks which are closer to those of the traditional art system in comparison to that of Hic et Nunc.

Keywords: crypto art, NFTs, artist legitimation, social aesthetics, platforms as intermediaries

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1. Introduction

The twentieth century saw the rise of Avant-guards, as well as social movements and transformations that profoundly changed the meanings, languages, and statutes of art. Probably the most significant change was represented by the process of dematerialization of art, which began in the 1960s with what Chandler and Lippard (1968) defined as "an ultra-conceptual art, that emphasizes the thinking process almost exclusively" (p.31). Towards the end of the century, the dematerialization of art in the new digital environments, which had been introduced by the profound transformations in information and communication technologies, an ideal ecosystem for further development. The invention of blockchain technologies determines yet a different environment that becomes a new field of experimentation for art.

The new disruptive and somehow ideological technological innovation determined the space and conditions for a new artistic phenomenon: that of NFTs (non-fungible tokens), one of the most innovative and discussed phenomena in the art world over the last decade. NFTs are cryptographic digital assets that, once created (*minted*, in blockchain language) exist on a blockchain (a peer-to-peer decentralized network where transactions are recorded in a distributed digital ledger) with unique metadata and identification codes that are representative of digital creative or artistic (and potentially also physical, not only immaterial) works.

Among their many applications, the one experienced in the art market allowed the flourishing of a new environment, one that is exclusively digital, fueling the development of a new artistic genre, defined as crypto art, similarly to the cryptocurrencies it is exchanged with. Known or unknown artists, digital or traditional, with their own identity or disguising themselves behind pseudonyms, individually or as collectives, have ventured into this new form of art, one in which popular NFT marketplaces act as gateways, mediating artist and audience interaction.

The new digital environment determined by NFTs forces us to redefine concepts such as ownership, individuality, and replicability; consequently, it questions the legitimation frameworks of artists and artworks, challenging the frameworks of traditional art itself. In the new century, the internet has already forced artists to learn how to dialogue with platforms and new digital languages, looking for the best ways to inhabit them, gain visibility and transform them into a sales and promotion tools. With NFTs, which introduce dedicated platforms and new mechanisms pertaining to their use as well as to the market, such an exercise inevitably becomes a new challenge that calls for new rules dictated by a different technological, social, and economic context which requires artists to better define themselves.

We could say that crypto art represents, in the art world, what finance has represented for the economy, financializing the economic system. Indeed, the production of wealth through finance, instead of through industry and classical economics, is similarly reflected in art: art has followed these dynamics of socio-economic transformation, dematerializing itself in the online spaces of a new digital economy, raising those

concerns that see NFTs capable of completely transforming art into a speculative financial instrument (Price & Kuo, 2021).
If internet, with its networking potential, had already granted and forced the artist into greater autonomy, crypto art is starting a new phase, moving the horizon of digital art even further. As it develops through blockchain technology, crypto art brings with it the ideologies introduced by the underlying visions of crypto enthusiasts. Knittel and Wash (2019) define them as "True Bitcoiners" (p.3), a group of crypto enthusiasts who do not limit themselves to using or analyzing the characteristics of such technologies but recognize in these technologies the possibility for an ideological social change, based on the underlying principles of blockchain technology: “persistency, decentralization, auditability and anonymity” (Bsteh, 2021, p.5).

With the rise of digital art, starting with the development of information technologies and further fueled by the Web and its subsequent transformations, artists and activists play a fundamental role in terms of questioning the dominant modes of digital interaction and artistic independence. In fact, the underlying visions for NFT, with their liberation rhetoric, appear to promise artists more freedom, suggesting a world that is detached from the current mediated frameworks of the traditional art world, where legitimation mechanisms seem to be determined by a limited number of dominant voices and prominent entrepreneurs in a social context (Crane, 1976): “the digital and contractual determination of crypto art […] allows for the hyperportability of the object and the rejection of conventional markets and institutions” (Franceschet et al., 2021, p.403). In this dimension, the artists engaged in the production of digital art or in bringing their artistic production into a digital dimension, must deal with the new market (while the other markets continue to be active: the traditional market characterized by material exchange, and the former digital one), also introducing their own original rules of demand and supply in relation to those that have governed the art market in the previous decades (Bsteh, 2021). In such regard, new technologies exasperate the artist’s autonomization process that began, as Bourdieu revealed, many centuries ago, when art started freeing itself from its dependence on the powers of the church and patreons (Bourdieu, 1985).

Today that autonomy could be compared to a fragile loneliness if seen in relation to the market and the extreme competition that rules within these new environments. What is more likely, though, is that there has simply been a change of setting, where the church and the nobles of the past have been replaced by a more complex system and by less recognizable subjects, interested in controlling the profit logic that rule over much of the art system. As mentioned by Bourdieu (1985), “The end of dependence on a patron or collector and, more generally, the ending of the dependence upon direct commissions, with the development of an impersonal market, tends to increase the liberty of writers and artists. They can hardly fail to notice, however, that this liberty is purely formal; it constitutes no more than the condition of their submission to the laws of the market of symbolic goods, that is, to a form of demand which necessarily lags behind the supply of the commodity" (p.16). Either way, this form of autonomization, later fueled in the contemporary digital era with artists increasing autonomy and detachment from mediators pushes artists to become self-entrepreneurs, developing

a new inclination that belongs to entrepreneurs (Barry, 2011; Essig, 2015; Kackovic & Wijnberg., 2022; Poorsoltan, 2012; Scherdin & Zander, 2011).

Just as the world of contemporary art has seen a conflict between the real and digital world, when online art marketplaces were able to present an offer that, only a few decades ago, could be exclusively found in art fairs and galleries, so too the development of blockchain technology has required the creation of ever new and different platforms to respond to the new digital needs of art. Crypto art needed dedicated platforms, in order to better exploit and ease interaction with blockchain technology, where artists, audiences, and collectors meet. These platforms, like the art platforms of the pre-blockchain era, perform the function of mediators as they mediate between user interaction by hosting, organizing, and storing but also circulating the content produced by its user base (Gillespie, 2017). The mediation between artists, audiences and the market is never neutral, if only because it involves a process of legitimizing the art and the artists it proposes. Legitimation does not only affect and involve the artists, but it also participates in determining the identity and leading values that will define the boundaries of a new system. In other words, all the subjects that publish, interact, navigate, and trade in the platform will determine the direction of this new field.
If, in the traditional art system, the legitimation processes are conditioned by many subjects — such as publishers, critics, art dealers and traders as well as other competing dominant voices (Bourdieu, 1980) — the systems of legitimation now proposed by these new platforms apparently reduce the influence of gatekeepers and detach from established institutions (Bsteh, 2021).

While being aware that the scenario concerning NFTs is particularly dynamic and its statutes are still in constant evolution (a condition that also influences ongoing or future modifications in the current research topic), this research aimed to examine different models proposed by platforms explicitly dedicated to NFTs exchange. In particular, it will focus on the ways in which NFT platforms determine processes of legitimation, establishing the rules and social targets they aim to reach. How do NFT platforms legitimize the encounter between the artists and the audience and the mechanisms of value exchange between them? To what extent do these platforms promote different and alternative systems of artist legitimation compared to those traditionally used by the art system?

This thesis aims to analyze how the systems of presentation and diffusion of crypto art are mediated as well as reflected within the social contexts that NFT platforms promote for artists engaged in the production of crypto art.
Rooted in the debate on artist autonomy, visions introduced by blockchain technology and the ways in which platforms act as intermediaries, this thesis seeks to describe and analyze some of the latest changes that are challenging the world of art in terms of alternative legitimation systems distant from conventional markets and institutions (Franceschet et al., 2021)

This thesis studies platforms that use blockchain technology, being aware of its limits, given that "decentralized networks create a discovery problem, where you can't find information if you don't know where to look." (Gorcenski, 2022, para.3).
The research does not ignore the ways in which blockchain technology and its underlying ideologies (leading a shift to Web 3.0) are manifesting and sparking debates along the lines of Web 2.0's heavily centralized paradigms (Turner, 2006; van Dijck et al., 2018). While blockchain technology appears to address some of the challenges that characterize Web 2.0 (Gorcenski, 2022), there seems to be a conflict between the promises of Web 3.0 (the future that maybe blockchain will participate in defining) and the problems of Web 2.0. As it is true that blockchain is a decentralized technology, it is also true that it manifested in the development of highly-centralized platforms, which respond to logic that manifestly finds itself in contradiction with the underlying decentralization ideologies and visions of crypto and NFT enthusiasts.

The research analyzes the platforms of Opensea and Hic et Nunc. As Opensea emerged as the first online NFT marketplace (built on the Ethereum blockchain) and is widely regarded as one of the leading players in the NFT ecosystem, the platform of Hic et Nunc (built on the Tezos blockchain), in only two months from its date of launch, has quickly surpassed the number of transactions made on Opensea. However, only eight months later, Hic et Nunc was shut down by its creator without providing any reasons, leaving to the community its only hope for its resurrection. In such regard, after its shutdown, it has seen the contribution of many reworks of the platform from anonymous developers and programmers who wanted to keep alive the visions for a more decentralized ecosystem within the NFT space, far from the typical intermediation roles of corporate and commercially oriented platforms. Its impact was so significant that ultimately its community joined forces and regrouped in a fully decentralized new platform, named Teia. In the awareness of the vision it sparked, although the original platform of Hic et Nunc ceased to exist, this thesis aims to confront the platform of Opensea to the original platform of HEN during the peak of the NFT phenomenon in 2021.
Other than being two of the leading players in the NFT space, the two platforms have been selected for the scope of this research as they represent two opposing extremes in the online NFT marketplaces; the first is characterized by a corporate statute and based on a mainly economic dimension, the latter is independent and community-driven, that instead favours a non-speculative dimension. As the thesis will demonstrate, these two models define and characterize the legitimation structures that represent two social contexts that — differing in values, visions and ideologies — polarized more than any other platform the audience that made NFTs a global success. Involving two very different types of audiences, such polarization confirmed these platforms' statuses and conferred legitimacy to the artists who inhabited them.
In the consideration of platforms as sites of analysis, the research examines the governance models they are based on, as well as the artworks and the presentations of the artists they contained adopting an approach to social aesthetics.
For the first section, the thesis will consider the artistic creations that are presented in the two platforms, intended as products of the cultural and social contexts these spaces promote. Art objects influence and model

the cultural environment and, in the same way, adapt to the cultural environment they inhabit (Moustaira, 2011). The following section will address the presentation of artists. It will focus on the way artists present themselves on the two different platforms, as well as the way they present themselves on Twitter, the platform preferred by crypto-artists and NFT enthusiasts to interact in the digital space. The final section of the analysis will identify the governance frameworks suggested by the two platforms, in relation to the models each platform has selected to define the rules and processes that promote or limit the artists' and, more in general, the users' participative and interactive mechanisms. The thesis will argue that among these two different social contexts, that of Opensea, appears to propose legitimation frameworks that are closer to the economic and profit-oriented conception of art as commodity of the traditional art system in comparison to the platform of HEN.

In the context of the analysis that follows the thesis will refer to the HEN platform using the present tense, as if at the time of the comparison with the Opensea platform it was active (which actually happened between March and November 2021) and had not undergone the sudden suspension in November 2021. This decision was deemed necessary to ease both readability as well as the development of the discussion.

1.2 Opensea and Hic et Nunc, the Two Platforms Analyzed

OpenSea is a platform created in December 2017 for exchanging and creating NFTs. It was the first market for non-fungible tokens, the first platform that gave the possibility to buy crypto collectibles and exchange art and digital objects with cryptocurrency. It initially supported the Ethereum blockchain, but then also included the Polygon, Solana and Klaytn blockchains.

In just a few years it has become one of the most popular, hosting some of the most economically valued projects in the NFT space such as "Crypto Punks" and "Bored Ape Yacht Club". Most of these popular projects aim to attract users by promoting their NFTs as providing a value of utility that comes from ownership of these digital assets, as users can use them to authenticate their identity across a variety of digital spaces. Anyone can access and view the works of crypto art and other NFTs in the catalog, all it is needed is a web3 wallet and cryptocurrencies (payments can be made in 240 different cryptocurrencies). Opensea is one of the protagonists of the social phenomenon and the extraordinary economic overvaluation of NFTs which have also become a media phenomenon, introducing world public opinion to blockchain-distributed digital art in 2021.

Hic Et Nunc, also known as HEN, was launched in March 2021, four years after Opensea, by a programmer named Rafael Lima. Unlike the platform of Opensea, Hic et Nunc is an open-source platform, where all the smart contract infrastructure, its modules and software components were made available on a dedicated publicly accessible repository on GitHub.

Being built on top of the Tezos blockchain, NFTs are also referred to as OBJKTs, terms specifically used for NFTs minted in the Tezos blockchain. Interacting with Tezos, it took on a more environmental and economic

identity than Ethereum's popular Opensea platform, whose commission costs for each transaction are much more expensive.

Hic Et Nunc offers artists and buyers an economically more convenient platform to use. Just as for the platform of Opensea, when a transaction is successful, the commission to be paid to the platform corresponds to 2.5% of the total amount, which in the case of HEN is instead paid directly to the creator for its maintenance.

The platform's native utility and governance token named hDAO, allows it to be used both as a "utility token" (hdDAO coins with which one can upvote artworks, to promote NFTs in the homepage's default search filter), and as a "governance token" (however this second function was never accomplished and the platform never stated how the "democratic tool" could have been used towards a full decentralization).

Although its immediate success, it did not last for long: opened in March 2021, in November of the same year its closure was announced on Twitter, without explanation. Artists, programmers, specialists, owners who had participated in the few months of HEN's life, tried to rise again the platform, demonstrating that in those few months it had built a solid community that embraced the visions of its creator.

2. Theoretical Framework

2.1. From Patrons to Artistic Autonomy: the Legitimation Frameworks of Art

Throughout the Middle Ages, artistic and intellectual activity was controlled by external sources of legitimacy and mainly served the function of social servitude. It was only later, with the cultural revolution started with the Renaissance, that it slowly started freeing itself from the powers of aristocracy and the church. (Bourdieu, 1985) Bourdieu affirms that this process of detachment from their dependence on church and aristocracy is correlated with the diffusion of primary education leading to the rise of audiences intended as consumers of culture alongside an increasing social differentiation, guaranteeing artists more autonomous frameworks of economic activity as well as preparing the ground for a competitive environment for acquiring cultural legitimacy (Bourdieu, 1984). Bourdieu additionally suggests how this period is also correlated with "the constitution of an ever-growing, ever more diversified corps of producers and purveyors of symbolic goods, who tend to reject all constraints apart from technical imperatives and credentials" which he identifies as "a socially distinguishable category of professional artists or intellectuals" (Bourdieu, 1985, p.15).

Throughout this process of detachment from aristocratic and ecclesiastical powers, the author describes a process of distribution of the consecration power of authority, where academies and salons, as much as institutions, publishers and other external actors are involved in what he describes as a selective process for cultural legitimacy (Bourdieu, 1985). Within these more autonomous frameworks (accelerated by the significant transformation processes started by the industrial revolution and later continued during Romanticism) artists and intellectuals progressively start to liberate their works from their traditional social function, described by Bourdieu as "the moral censure and aesthetic programmers of a proselytizing church or the academic controls and directives of political power". (Bourdieu, 1985, p.15)

Bourdieu (1985) defines the field of cultural production as a market of symbolic goods in which complex and interrelated systems of institutions and agents define their role and position in the functions they serve (ranging from production to diffusion of symbolic goods). He defines a symbolic good as a good that can be understood simultaneously as a commodity as well as a symbolic object. In this perspective, the cultural and economic values "remain relatively independent although the economic sanction may come to reinforce their cultural consecration" (Bourdieu, 1985, p.16).

The increasing diversity of the audiences to which producers aim their products, (defined by Bourdieu as process of differentiation) among what he defines as "fields of practice" has led to the development of theoretical principles (on which the systems of economics, politics, law, art have developed) which ultimately are defined in the same differentiation of the social structures, underlying ideologies and value spheres by which they are constructed. In this way, the gradual development of art as commodity with the emergence of a distinct group of producers competing in the market of symbolic goods, by detaching from a conception of

art as commodity from that of pure symbolism, opened the way for what the author defines as a pure theory of art: an "art as art" (Bourdieu, 1985, p. 16).

The progressively autonomous artistic environment, freed from patrons or collectors throughout the development of an impersonal market, on one hand, increased the liberty of artists and intellectuals on the other, it represents their condition of submission to the frameworks of the market of symbolic goods, governed by the commodity's laws of demand and supply (Bordieu, 1985). In such regard the author affirms that "they are reminded of this demand through the sales figures and other forms of pressure, explicit or diffuse, exercised by publishers, theatre managers, art-dealers" (Bordieu, 1985, p.16).

In discussing artists' autonomization, Bourdieu (1985) stresses the role these secondary actors play in determining which artists and artworks are diffused or presented in social spaces. It is exactly within these mediated frameworks that to enter in competition on equal footing with incumbents of the market, artists must at least gain a state of proper legitimacy for surviving as an entrepreneur: "a state reached […] when a source carefully assesses a new entrepreneur's propriety and then makes a judgment about the appropriateness of that entrepreneur in a social system." (Kackovic & Wijnberg, 2020, p.1093).
Kackovic and Wijnberg (2020) investigating the legitimation frameworks of the primary market for contemporary visual art stress that the differentiation and credibility of these third-party sources evaluating cultural products play a central role in influencing mediators' decisions (referring to art galleries) in forming and conferring legitimacy but also granting new entrepreneurs full-fledged entry into a focal market. In fact, the process of a mediator's decision to grant proper legitimacy to a new entrepreneur does not limit itself to the relationship between the artist and the respective gatekeepers.

Similarly, Crane (1976), following Bourdieu, suggests that as mass media critics influence a mediator's decision in legitimation processes, their function as gatekeepers is joined and constructed together by incumbent entrepreneurs as well as the dominant voices accountable for the diffusion of art within a social context. In such regards, also the artists themselves competing in a market can be understood as contributing to the determination of frameworks of legitimacy: the artistic context and environment in which artists are competing themselves is determinant as their competition develops in claims to orthodoxy, "to the legitimate and monopolized use of a certain class of symbolic goods" (Bourdieu, 1984, p.7).

While legitimacy can manifest at different levels (Becker, 1982) and can originate from a wide variety of sources in different ways (Crane, 1976), Zelditch (2001) describes legitimation as a process through which the unaccepted becomes accepted through consensus. This process occurs if it aligns with the rules, values, practices and beliefs endorsed by a specific social group (Zelditch, 2001). However, consensual acceptance in a social context cannot be absolute. As complete consensus within a social sphere is hardly ever reached it is

necessary to define this term more specifically: consensus necessarily exist at the level of a social group and not necessarily at a singular one (Zelditch, 2001).

It appears clear, as Bourdieu suggests, that the process of autonomization which has contributed to arts' liberation towards a progressive artistic independence operates within market-driven dynamics and complex structures and processes of legitimacy that govern the art world. In light of arts' autonomization and the increasing orientation towards an impersonal market, the freedom of artists, as argued by Bourdieu (1985), appears to be "purely formal".

2.2. Dematerialization of Art's Aesthetics

In contemporary art, the idea, concept and experience that constitute an art object matter more than the actual materiality of the artwork itself. Pascal Unbehaun (2021) in such regards suggests how the meaning has progressively changed as among the more traditional form of art projects, also artistic research, interventions or social practices have started to be considered art.

These phenomena have been addressed by many scholars engaged in art discussion (Clark, 2010; Lippard, 1997; Napier, 2015) as the dematerialization of the art object over the last century, which developed in parallel to an increasing autonomization of artists.

Although this term is commonly associated to the 1990s with the development of Net art, its meaning and definition reach over the simple aspect of non-materiality: it is part of a progressive cultural process that started with the Avant-garde and was further fueled by the conceptualism of art. Coined in 1968 by Chandler and Lippard, this term describes the state of conceptual art in the 1960s: an art that focuses on the concept and thinking process underlying the creation of art, rather than the actual object produced.

While during the beginning of the 20th-century abstract art, which traces its roots back to the late 19th century, was engaged in trying to introduce a new perspective promoting new levels of appreciation outside of the traditional canons of objectivity introduced by Realism of the avant-garde, conceptual art, which emerged towards the end of the 1960s, begins to challenge the concept of the artwork, introducing a view of art where concepts and ideas of the artist take precedence over artwork's materiality and traditional aesthetic concerns. Contemporary art, especially after the 1960s, has seen repetition as one of the most diffused methods of expression, and artworks started being considered as a metalanguage that reflects on the same artworks as much as on popular culture.

As artistic expression starts to challenge traditional frameworks, art starts to orient towards a direction that can be understood as a game of messages, in which the artist invites the observer to a conceptual dialogue: "as if

it was a private joke between artists, the wealth of reproduced and legitimated art, and the new public, knowledgeable of a pop expressiveness and a media culture that is increasingly dense" (Queiroz, 2017, p.1).
It is in the midst of these radical changes in the traditional aesthetic paradigm characterized by conceptual art that the ground is prepared for the rise of new media art: artists start seeing a liberatory potential in the networking and communicative allowances of technologies being introduced towards the end of the twentieth century.

The initial phases of new media art were characterized by experimenting with new tools and mediums to produce, modify and disseminate artworks.
The development of computers in the 1960s and their progressive popularization becoming tools used in everyday life through the increased social dynamism introduced by Web 2.0, increased the accessibility to query and distribute images, experience and reproduce art.
The internet years represent a period of radical transformation in digital art's history and engagement. These networking and communicative technologies further fueled the expansion of producers and audiences by "accelerating the production, delivery, and consumption of goods and services" (Sidorova, 2019, p.2). With the development of information and communication technologies, art has developed in many directions and due to this expansion of artistic expression through digital technologies. For such it appears hard to define digital art within specific boundaries.
Digital art, initially called new media art or computer art, can be used as an umbrella term for what has been defined as "internet" and "post-internet" art, which implies the act of producing art using electronic devices. While from what was originally referred to as digital art, "internet" described more specifically art produced during the 1990s in Web 1.0, the term "post-internet" was later introduced to describe art produced in the gradual technological transition towards Web 2.0.

2.3. The Visions for NFTs and Potential of Blockchain Technology in Artists' Autonomization

The rise and later developments in the field of information and communication technologies (ICT) have significantly affected and transformed the economic and socio-cultural spheres of society (Samdanis, 2016). Generally speaking, as argued in the previous section, in the history of the evolution of digital art, networking and communicative technologies have contributed significantly to increase art's accessibility and artists' freedom, challenging the state of art. The developments of Web 1.0 introduced in networked spaces a variety of tools to simplify processes to query and organize information. From search engines, knowledge repositories and static web pages of Web 1.0, the web started to orient towards the more dynamic digital environments of Web 2.0, dominated by platforms (Sykora, 2017).
Former hacker Kevin Mitnick (2010), in regards to the current state of social spaces where the social attention a digital object attracts (such as the shares, followers, likes, and comments) appear to be more valuable than

the actual message of the content transmitted, describes social engineering skills comparing them to performative arts: those of persuasion and influence.
Similarly, Kang et. al (2019) in regards to the potential of digital technologies (particularly with Web 2.0) to transform the current state of art, suggest how the same value of art has progressively changed: “digital interaction with the audience can promote cognitive decoding and enhance kinesthetic and emotional responses to artistic activities” (p.3).
In the perspective of the increasing autonomy of artists in the modern era, the idea of the artists has been increasingly associated by many scholars engaged in discussing the state of art with an entrepreneurial stance (Barry, 2011; Essig, 2015; Poorsoltan, 2012; Scherdin & Zander, 2011). Kackovic and Wijneberg (2020), building on this perspective, focusing on the primary market for contemporary visual arts, discuss about artists as self-employed entrepreneurs suggesting how they tend to display traits of innovative and less influenced artistic orientation as well as being more inclined to an attitude in opposition to the traditional frameworks of art.

While art started to progressively dematerialize (Clark, 2010; Lippard, 1997; Napier, 2015), detaching from its sole manifestation as an object, its dematerialization developed in parallel with the digitalization of the economic system.
The dematerialization of the economic-financial system, along with globalization, is one of the identity traits of the neoliberal era inaugurated by Margareth Thatcher's (1987) famous statement "there is no society". Art has followed these dynamics of socio-economic transformation, dematerializing in the online spaces of the new digital economy, to the point of becoming itself a speculative financial tool (Price & Kuo, 2021). In this digital economy, far from a conventional one and much more volatile, there appears to be a strong link between the value of an asset and its social status (Howcroft, 2022).
After all, what else could one expect from art if even the monetary system was adapting to the new paradigm, dematerializing into cryptocurrencies?

From what has been discussed by technological activist Emily Gorcenski (2022) as a shift from the decentralization of information of Web 1.0 to the decentralization of culture of Web 2.0, the Web ecosystem has been subject to an increasing rise of the influence of digital platforms within the cultural industries. This process has been described by Nieborg & Poell (2018) as platformization of culture, which more specifically refers to “the penetration of economic and infrastructural extensions of online platforms into the web […] fundamentally affecting the operations of the cultural industries” (p.4276).
As this perspective finds its conception in relation to Web 2.0 it also finds relevance in the perspective of blockchain-based platforms of Web 3.0, where in the case of NFTs, marketplace platforms act as gateways to production and access to digital art. Along the lines of the idea of dematerialization of art in light of crypto art and the potential offered by decentralized technology, Franceschet et al. (2021) affirm: “the digital and

contractual determination of crypto art […] allows for the hyperportability of the object and the rejection of conventional markets and institutions" (p. 403).
However, as the decentralization of the Web's services has not been challenged as much as they have until now with blockchain-based applications, issues linked to Web 2.0's critique of centralization inevitably seem to arise also in Web 3.0's landscape as "decentralized networks create a discovery problem, where you can't find information if you don't know where to look" (Gorcenski, 2022, para.3). This controversy repurposes in the current landscape of crypto art online, where artists disseminate their works through NFT marketplaces to list digital objects which exist within the frameworks of these platforms' search filters and users' navigation and interface designs.

Crypto art is one of the most recent artistic movements, where the artist engaged in its production creates "works of art, typically still or animated images, and distributes them via a crypto art gallery or their own digital channel using blockchain technology" (Franceschet et al., 2021, p.402). This artistic movement is tied to the technological mediums in which it exists as much as to the values (and underlying ideologies) that blockchain technology has cultivated (Franceschet et al., 2021).
Just as post-internet art in its manifestation reflects a consciousness of the same networks it inhabits - from the initial conceptualization phases of its production to the audience's reception - (Archey & Peckham, 2014, p. 8) these ideologies reflect in the same aesthetics of these artworks as much as in the artists' presentation in marketplace platforms and social spaces where artists share their works and interact with their audience: "art objects […] adapt to the cultural environment they inhabit" (Moustaira, 2011, p. 4).

The rising visions for art introduced by NFTs and the potential of blockchain technology applied to the digital art field with its underlying decentralizing and democratizing precepts can be understood along the lines of this process of expansion of artistic autonomy which was fueled by Web 1.0, with its promises to decentralization and democratized sharing of information (Gorcenski, 2022). In such regard, SuperRare's gallerist Jonathan Perkins describes crypto art as "a new idea in an ancient field" (Franceschet et al., 2021, p.404). Along the visions that fueled the initial phases of the web, crypto enthusiasts recognized the potential of decentralization represented by blockchain technology. In particular, blockchain technology was constructed along four main principles: "persistency, decentralization, auditability and anonymity" (Bsteh, 2021, p.5).
Regarding crypto ideologies Knittel & Wash (2019) affirm how this movement seems to be based on an ideology about social change identifying in a discourse research on the Reddit platform around the topic of crypto and bitcoin, a group they define as "True Bitcoiners" (p.3), which rather than talking about blockchain technology or specific uses or applications of cryptocurrency, was engaged in discussions that revolved around the topics of trust.

In crypto art, the themes of decentralization are themes that develop from the viewpoint of artists, collectors and also from actors acting as mediators in the traditional art market that are following the evolutions of an art world which is never static. As for the artist, crypto art represents "a way to get and keep control of their artworks and reap related benefits" (Franceschet et al., 2021, p.404) on the other hand, it inevitably implies a higher degree of autonomy which, besides the new potential of decentralized technology introduced by blockchain, had already started with Web 2.0's social media reshaping the same role, engagement and notion of the artist.

2.4. Governance of Platforms: Crypto Art Platforms as Mediators of Crypto Art

From the dematerialization of art and the increasing digitization of objects, the NFT space dominated by platforms that mediate users' interaction with the blockchain can be understood along the lines of the process of platformization of culture, suggested by Nieborg & Poell (2018). In describing this phenomenon, Nieborg & Poell (2018) affirm: "cultural content producers have to continuously grapple with seemingly serendipitous changes in platform governance, ranging from content curation to pricing strategies" (p.1).

While the mediators of art were institutions, museums and galleries which acted as gateways influencing artists' gaining of legitimacy, in the space of crypto art (for what concerns the digital space) the same position is now substituted by platforms.

It appears relevant to understand the dynamics that regulate frameworks of legitimacy in these art platforms as crypto idealist visions originate from ideologies of reshaping traditional centralized structures of governance, which dominate and mediate the current frameworks of Web 2.0. In the same way, the visions for the application of these technologies by crypto artists, reflect the idea of decentralization and independence from mediation, far from the traditional art world of conventional markets and institutions.

As suggested by Plantin et al. (2016), as much as platform studies focus on investigating the participatory mechanics that characterize media practices, they also more directly address the double-faced reality of commercial platforms: "their affordances support innovation and creativity […] yet simultaneously constrain participation and channel it into modes that profit the platform's creators" (p. 297). In this perspective, Langlois and Elmer (2013), providing the example of Facebook as a platform of corporate nature, underline how the economic interests of platforms affect the interface designs through which they mediate and determine mechanisms of social interaction.

Bucher and Helmond (2018) instead have critically analyzed platform design reflecting on the concept of affordance through which to understand and analyze social media platforms. Affordances of platforms offer a perspective through which to comprehend their roles as mediators as they standardize modes of engagement (Bucher & Helmond, 2018).

At the same time Gillespie (2017), in discussing the governance of platforms, argues that the current frameworks of dominant platforms in their claims to impartiality and in freeing information and expression are only apparent, as ultimately "everything on a platform is designed and orchestrated" (p.4). In such regard, Gillespie (2017) suggests how modern platforms' mediation manifests in institutionalized, proprietary and opaque ways, becoming "public arbiters of cultural value" (p.14).
In light of platforms' critiques and their role as mediators, many scholars in the field of media studies have focused on investigating the frameworks of governance of platforms, focusing on the duties and corporate practices of many popular modern platforms (putting a major focus on social media platforms) understood as dominant intermediaries of contemporary media (Gillespie, 2017; Van Dijck, 2012; Zhang et al, 2022). In such regard, Zhang et al. (2022) defines platform governance as what "concerns the design and deployment of governance choices, including decision rights, incentive structures, and control mechanisms" (p.602)

While the term governance in media studies is usually associated to modern Web 2.0 platforms, when related to NFT platforms of Web 3.0 it assumes a more complex meaning, due to it being a fundamental characteristic that defines the underlying technology with which these platforms constantly interact (Fischer & Valiente, 2021).
In blockchain technology, the term governance can be used to describe two different contexts: it can be referred to the blockchain itself — intended as the set of changes of the software that govern the blockchain and the rules which determine its functioning — or it can be related to the use of blockchains to govern — such as the platforms or other services which are to some extent blockchain-based (Fischer & Valiente, 2021). In the second case, this term has been increasingly linked to what concerns Decentralized Autonomous Organizations, also known as DAOs: decentralized governance models in which the shareholders determine how to allocate the organization's funds and modify its code (Buterin, 2013).
Overall, the aspect that mainly characterizes this technology relies on the models of distributed governance it enables, in a system where the ability to govern is distributed among all of its participants: "the means of achieving the direction, control, and coordination of stakeholders within the context of a given blockchain project to which they jointly contribute" (Pelt et al., 2021, p.1).

Although, as argued by Murray et al. (2022) the evolution of the web towards what has been defined as Web 3.0 seems to be creating an internet that relies less on centralized firms which mediate users' activity, as argued in the previous section, it also appears how platforms, easing users' interaction through designed interfaces, inevitably play a similar role of intermediaries as for platforms in Web 2.0, which repurposes on the design as well as governance models.
In this understanding, within the NFT ecosystem, where artists compete for legitimation, platforms determine the rules to which artists and collectors must adhere, directly mediating interaction between artists and the audience. While these platforms interact with the blockchain mediating the exchange of crypto art, the desires

and underlying visions of decentralization of NFT enthusiasts and crypto art rooted in blockchain technology, appear to inevitably find frictions with the current centralized frameworks of Web 2.0.

3. Methodology

3.1. Introduction

The thesis aims to shed light on the legitimation frameworks that emerge in the HEN and Opensea platforms. Considering the different values they promote, the research will analyze the two platforms in order to interpret and process results that reveal the social structures characterizing the two different digital spaces. Using a qualitative phenomenological approach, the research will initially focus on the artist-platform relationship to highlight the values and visions within which such a founding relationship develops.
This will allow us to analyze the mediation and interaction that platforms offer to the relationship between artists and the public. The relationship models (artist-platform and artist-audience mediated by the platform) constitute the system that justifies the existence and activity of the platform itself and, consequently, inevitably reflect the identity of the two different social spaces, allowing differences and analogies to emerge.
Considering how the academic debate in platform studies has focused on the duties and implications of the most popular modern platforms, understood as dominant intermediaries of contemporary media (Gillespie, 2017; Van Dijck, 2012; Zhang et al., 2020), the present analysis will investigate artworks, artist presentations and governance of these platforms in light of the role they occupy in contributing to determine the most recent phases in the development of the NFT phenomenon, focusing on the frameworks of legitimation they promote. In particular, the research aims to understand how the phenomenon of NFT evolves and modifies the traditional statutes of the art world through an aesthetic analysis of the platforms of Opensea and HEN, due to the adhesions of the public and artists and their respective value settings that appear to represent different, if not opposite, concepts of platform.

In this research, the analysis of the artworks, of the artists and of the ways in which artists present themselves to the public, is completed by (also because it is interconnected to) the observation of the governance models proposed by the two platforms, interpreted in the light of the existing theory.
The objects and contents analyzed for the study of the artworks and the governance models were extracted from the examined platforms. As for the ways in which artists choose to present themselves and their poetics, these develop not only in the type of functions that the platforms allow artists to use but also through the many digital channels that offer artists the opportunity to multiply and specify their personal style of communication. Therefore, this part of the analysis related to the artists is, not limited to the observation of their presentation within the platforms, but also extends to the observation of consistencies and possible inconsistencies that can be detected in their activity on the social platform most used by crypto artists: Twitter.

While for Opensea, the analysis was conducted directly through the original platform's website which has been operative since its launch in 2017, for the analysis of the HEN platform, after its recent shutdown in November

2021, the research utilized one of the first website's reconstructions that was built from the platform's public GitHub repositories. Created by an anonymous developer that goes by the name of "Manticort", who helped Lima maintain the original website as a backup plan in case anything would have happened to HEN, users could access the platform again through an exact recreation of it.
As is the case with the other reworks of the original platform made by anonymous developers of the Tezos community, this mirror link (http://hen.teztools.io) was made possible by the fact that all OBJKTs remained in the smart contract ledgers. The codes of these smart contracts were made public on HEN's dedicated GitHub page, and the metadata linked to them were stored in the decentralized file sharing system on which the platform was built, utilizing the IPFS protocol.
In such regard, although aware of the uncertainty of its future state, this readaptation of the original HEN website was chosen, among the others, because it represents an exact copy of the platform: one of the few which maintained the front end (from the search filters to its design) as it was originally in the first version of Hic et Nunc. For such, the selection of this mirror website was made to allow for a detailed analysis of the platform as it was originally conceived in its operational status, before its shutdown in November 2021.
The indications gathered through the observation and the interpretative and comparative analysis of these two different social contexts are filtered and characterized by the social aesthetics methodological approach.

3.2. An Approach to Social Aesthetics

George Simmel, a German sociologist and philosopher, compared society to a work of art, an aesthetic phenomenon that should be observed both as aisthesis, a reflection on the perceptual experience of the world and as a philosophy of art or of cultural phenomena, which at the time he identified in advertising and fashion or as we could say today, design as well as social media platforms (Agnese, 2020). Today we could add digital environments, platforms and the design that characterizes their styles, their identities, their possibility for use and dialogue. "Society, argued George Simmel [...] is an aesthetic phenomenon where aesthetics is to be understood in its dual meaning: on the one hand, etymologically, [...] of reflection on the perceptual experience of the world; on the other, historically, of the philosophy of art or other cultural phenomena" (Agnese, 2020).

If Simmel, at the beginning of the last century, studied the dense web of mutual interactions between man, society and objects, calling the material he explored 'social aesthetics', today an analogous approach could concern the social spaces of platforms that, in the same way, create a dense interweaving of perceptions and interactions between users, objects (in this research intended as artworks) and the world.

At the center of this research is an in-depth study of the artistic, relational, selective, facilitation and governance characteristics that the two platforms offer to artists and audiences, not only as an initial cognitive analysis but also as an action capable of determining a development in the world. In fact, not only do platforms represent

the place where a mechanical exchange activity between artists and the public takes place, but also a place where social action is determined, which can produce cultural changes capable of designing, evolving, and transforming the collective and individual desire.
The relationship that binds the sociological to the aesthetics sphere allows us to understand the existence not only of a simple analogy between the world of society and aesthetics but to understand how mutual and interacting connections are established between the two areas. "It is only through aesthetic appreciation that the qualitative characteristics of social life can be recognized and studied as such" (Miyahara, 2014, p.1). In this regard, Welsch, while examining the phenomenon of global aestheticization, introduces a new concept of aesthetics that, in freeing itself from the work of art, proposes itself as an epistemological ground for investigating cultural and social contexts through sensory perception (Welsch, 1997). Helmut Staubmann (2022), in addressing the limits of sociology and social theory for the study of the quality of social contexts, identifies social aesthetics as an addition in redefining social sciences for investigating social, as well as socio-material, phenomena.

The three sections in which this research is divided (the aesthetics of the works of art, the aesthetics of the artists' presentation and the aesthetics of governance models) propose trajectories to re-interpret, also using an aesthetic-social approach, the modalities through which platforms not only mediate the direct artist-audience interactions but more generally produce new and much more complex social, economic and cultural interactions in the digital sphere. In this regard, it is sufficient to reflect on the cultural, economic and social implications that the NFT phenomenon has determined beyond itself, in the more complex social system of the art economy and in the productive and symbolic cultural exchange system, which appear with evidence in their more superficial dimension within the media debate that accompanied NFTs' rise and establishment.

In this perspective, an aesthetic-sociological reading of the use of the two platforms selected as sites of analysis (but more generally, of every platform) can help shed light on the evolving social contexts in which the legitimation processes manifest themselves as promoted by the two crypto art platforms, which might appear to be opposite, but actually take part and contribute to the same deep and complex transformation process of the digital art world.

3.3 Research Design and Data Collection

The research will complete the description of the selected artworks and of the examined artists, by providing direct documentation, through screenshots taken during June 2022, on the two platforms for the first, and also considering the social platform of Twitter for the latter. The same approach will be followed in the governance section, in which, in addition to an analysis of the governance models applied in the two platforms, screenshots as well as a discourse analysis of the platforms will be provided to aid the discussion.

The thesis will begin by examining the aesthetics of artworks as the first context for the interpretation of the two different environments represented by Opensea and HEN.

The artworks not only have value in themselves but because — in the aesthetic-social perspective — they participate in defining the social context: a) to which the platform wants to refer, b) that the platform intends to determine and create, c) within which the platform produces the terms of legitimation it offers to artists and their public, given that art objects influence and model the cultural environment (Moustaira, 2011) and understanding an epoch and its dynamics requires an interpretation through recontextualizing social theory to aesthetics (Staubmann, 2022).

This first section will analyze the aesthetics of artworks of each platform by highlighting the stylistic, compositional, cultural and technical differences that characterize some works of art that were chosen for their emblematic placement within the two platforms. In the great variety of artistic expression forms and styles present in the two platforms, according to the same navigational models promoted by the two platforms as indices of the popularity of artworks in these contexts, the research will focus on the top 50 ranking artworks, adopting a purposive sampling technique. Based on the two different models of navigation through which the platforms foreground artworks before others: a) for HEN (as a community-oriented platform), the analysis will consider the curated top 50 OBJKTs that appear in the "hDAO" filter feed of the landing page; b) for Opensea (as a highly profit-oriented platform) it will consider the 50 first collections classified as "Top" (repurposed in many sections across the platform) considering the “all-time” period frame, under the category of NFTs that the platform classifies as "Art".

The objects were selected during a one-month period in consideration of the fact that the rankings on both platforms would be subject to minor variations, which appear to be relatively insignificant in light of the scope of this thesis.

The screenshots of artworks selected among the “Top” collections classified as “art” on Opensea, in contrary to the “hDAO” top curated artworks of HEN, are captured to present not individual, but a series of the works pertaining to the respective collections analyzed. Considering the variety of formats that these platforms allow for, from moving images, 3d models to interactive artworks, the following analysis will present these creative works in the form of still screenshots, in case specifying instances of particular formats when necessary below the respective figures.

In the second section, the analysis will focus on the aesthetics of the artists' presentation, again examined in both platforms, this time extending the observation to the artists' use of social media to describe their profile.

The platforms of Opensea and Hic et Nunc offer artist-users the possibility of creating a personal profile in digital spaces. To allow a more complete interpretation of the values and social context that the artists of the two platforms represent and stand for, this section will be integrated with the analysis of the presentations with which the artists integrate their profiles in the relative platforms with Twitter. This social platform was chosen because it is the one most used by artists and collectors engaged in the NFT space.

As in the previous section, the artists analyzed will be selected among authors of the most acclaimed artworks and collections on the two platforms, on the basis of the different navigation and exhibition models "hDAO" and "Top" art that the two environments promote.

Finally, the governance aesthetics of the two platforms will make it possible to observe the models through which they operate as intermediaries, determining different practices and interaction mechanisms that each platform allows between artists and collectors, thus inevitably influencing the legitimation processes.

Considering the different governance models on which they have been built, an analysis of this specific aspect referred to the two platforms can reveal the ways in which it aligns itself with the promises of Web 3.0, and how each positions itself in the mediation role it has chosen to interpret in the crypto art space. Indeed, this section focuses on understanding these platforms' degrees of decentralization in terms of curating the structures, the moderators, and the regulatory practices that guide user navigation, and, in particular, the artists' visibility and autonomy in the platform.

4. Analysis

4.1 Aesthetics of Artworks

From an initial observation of the artworks, Franceschet's (2021) claims of crypto art's underlying technology which allows the rejection of conventional markets and institutions, inevitably appears also visible in the same aesthetics of artworks as "art objects adapt to the cultural environment they inhabit" (Moustaira, 2011, p.4). Considering the NFT space, it is no surprise that these artworks present themselves differently from what would be traditionally valued in the traditional art frameworks or from what would appear in an art gallery.
Upon first observation of the artworks appearing in the "Top" list of the "Art" category within Opensea, as well as of the top-ranking ones in the "hDAO" default search filter in HEN's Homepage, it seems how generally they both appear to reflect a wide variety of styles and forms, that also suggest diversified and non-hierarchical taste trends that the public has shown for digital art at this stage.
It is difficult to recognize trends within the very large catalog of works deposited in the two platforms, sometimes uploaded by the individual artists who then personally select the artworks to be exhibited in their personal profiles, sometimes by galleries, agents or collectives representing certain artists and acting as mediators, according to the values of their institution, organization or group. On the other hand, the different value settings that characterize each of the two platforms are inevitably reflected by these artworks and seem to have an influence on the characteristics of the artists that choose to appear on one rather than the other.

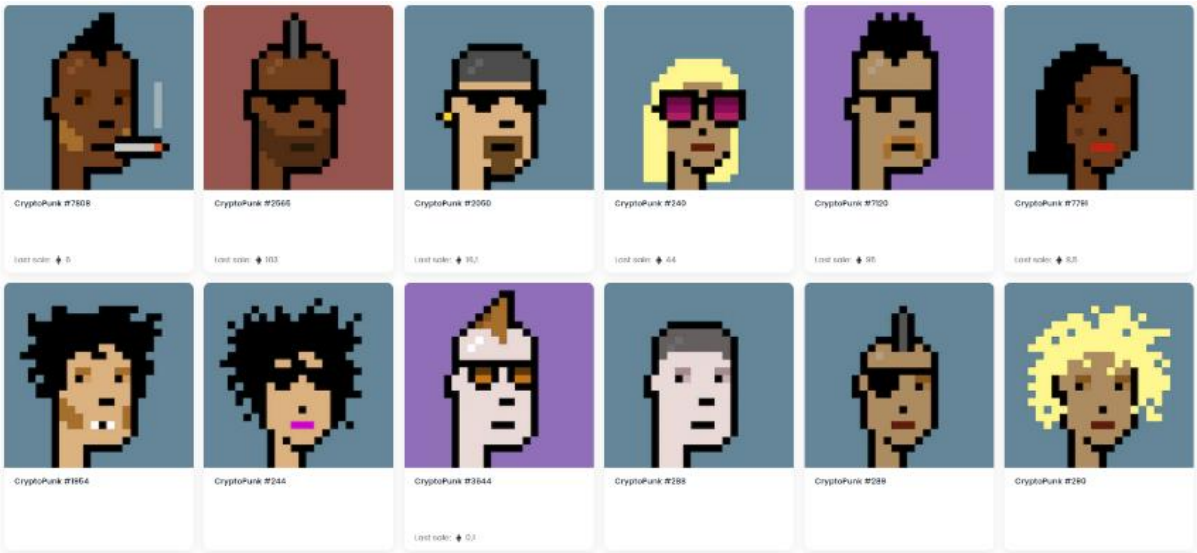
In crypto art, the works that are presented as series or as single works proposed in more copies (multiples), allow there to be more owners, who will be able to recognize each other, creating a potential community, able to share, in addition to the taste and pleasure for the chosen artworks, also subsequent dialogues and interactions across other digital platforms.
The collection feature in the platform of Opensea, through which artworks are presented in the platform's rankings, invites artists to engage in what could be defined as artistic projects rather than single artworks. Most popular collections in Opensea appear to announce and promise the owner future developments that are part of the virtual benefits of the purchase itself. Being an owner, therefore also entails being part of a community that has been chosen not only for similar taste and sensitivity. HEN, on the other side, does not include the feature of collections as in the case of Opensea, but only allows the upload of individual creative works, which can then be multiplied by the number of copies available for sale.
Considering that Opensea is a platform that privileges the economic dimension of the NFT phenomenon and, therefore, mainly dialogues with collectors and investors, its attention towards the creation of communities seems justified especially by the intention of making the sales proposal more appealing in the platform. HEN, instead, offers a selection of works that is not necessarily motivated by economic interests or composed of

collections necessarily thought for a market (or for an easier market), because it proposes a less market-oriented relationship between artist and audience.

While Opensea is characterized by a prevailing pop tendency, expressed in the retrieval of images and signs that belong to the art of more immediate consumption and that is appreciated especially by those generations whose taste has grown in the post-avant-garde period, the artworks in HEN seem to reflect a more experimental art, especially in terms of the artistic techniques adopted by the artists. This main difference appears evident as the majority of artworks placed in the "Top" NFT rankings of the "Art" category in this platform, present themselves as series of works, part of an artistic project, that respond to the logic of collections featured by the platform, frequently recurring to the use of repetition as main artistic means. This is the case of the top-ranking project "CryptoPunks", as shown in Figure 1, as well as the other successful NFT projects that follow the list, such as "Bored Ape Yacht Club" and "Lives of Asuna", as shown in Figure 2 and Figure 3 respectively.

Fig. 1

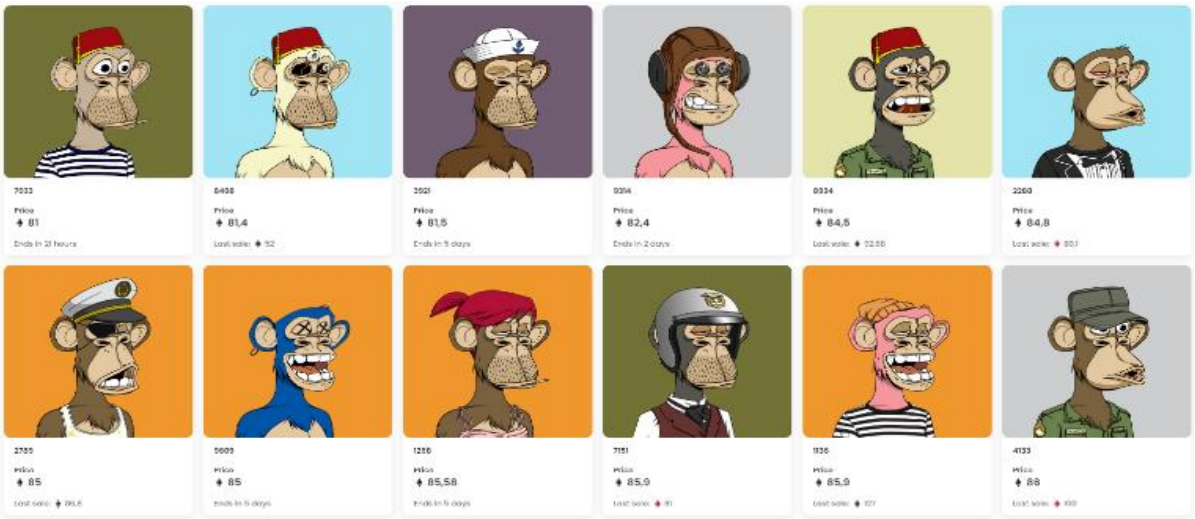
"CryptoPunks" collection by Matt Hall and John Watkinson, ranking in "Top" 50 "all-time" results of "Art" category on Opensea



Note. A series of works part of the "Cryptopunks" collection on Opensea.

Fig. 2

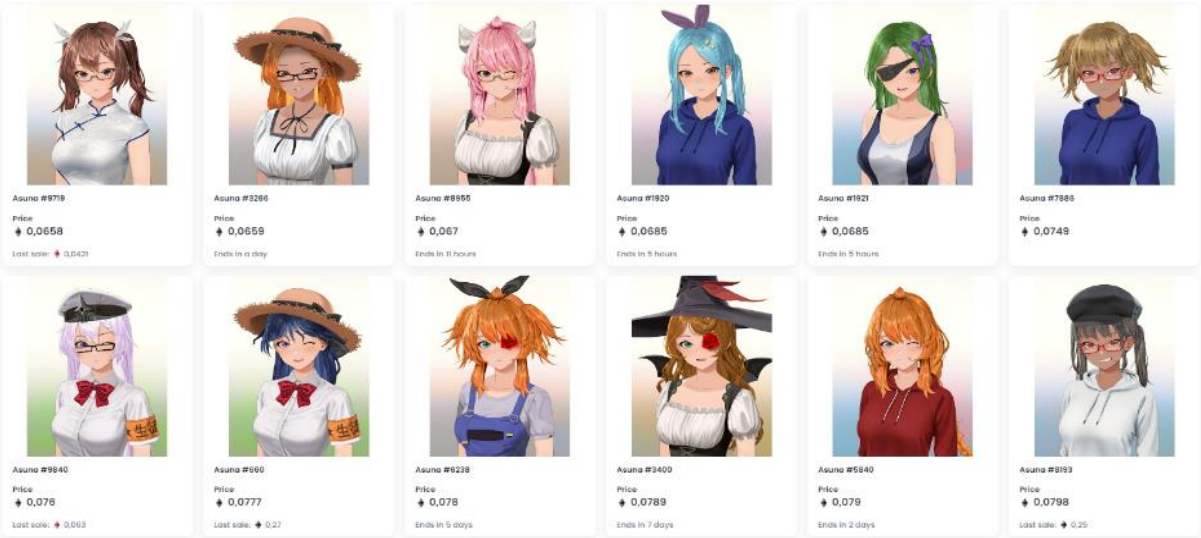
"Bored Ape Yacht Club" collection by Matt Hall and John Watkinson, ranking in "Top" 50 "all-time" results of "Art" category on Opensea



Note. A series of works part of the "Bored Ape Yacht Club" collection on Opensea.

Fig. 3

"Lives of Asuna" collection by Matt Hall and John Watkinson, ranking in "Top" 50 "all-time" results of "Art" category on Opensea



Note. A series of works part of the "Lives of Asuna" collection on Opensea.

As these collections on Opensea appear to be very similar in composition, representing characters and backgrounds that are arranged with the use of generative art digital techniques, in HEN the top "hDAO" curated artworks do not seem to follow such a strict compositional form. In fact, HEN appears to prefer a more refined stylistic research in the variety of forms, styles and techniques used, which sometimes even recalls already legitimized styles or even artworks that appear as background or setting to their subjects' actual artwork. Among the top 50 ranking artworks in HEN's "hDAO" category, as shown in Figure 4 and Figure 5

respectively, “Inu x Family - Daughter” by oshtru and “Window Still life 102” by john karel are examples of this tendency, recalling elements of classical art, such as, respectively, the colour fields of Piet Mondrian and the still-life genre. Other works, instead, recall elements of already acclaimed art in the NFT space, as is the case of “mtdapes #19” by mtdapes.tez shown in Figure 6, that clearly references the popular “Bored Ape Yacht Club” collection.

Fig. 4

“Inu x Family – Daughter” artwork by oshtru, ranking in top 50 “hDAO” results

Note. From Hic et Nunc, by oshtru, 2021, Hic et Nunc (https://hen.teztools.io/objkt/28134). Copyright 2021 by oshtru

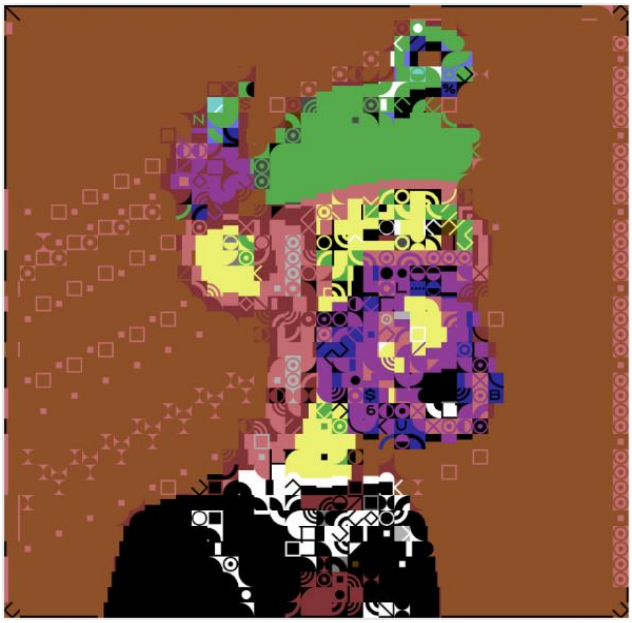
Fig. 5

“Window Still life 102” by john karel, ranking in top 50 “hDAO” results

Note. GIF. From Hic et Nunc, by john karel, 2021, Hic et Nunc (https://hen.teztools.io/objkt/56412). Copyright 2021 by john karel

Fig. 6

“mtdapes #19” by mtdapes.tez, ranking in top 50 “hDAO” results



Note. From Hic et Nunc, by mtdapes.tez, 2021, Hic et Nunc (https://hen.teztools.io/objkt/34711). Copyright 2021 by mtdapes.tez

Among the works that re-elaborate very recent NFT works, treating them as if they were art classics to be reinterpreted, "mtdapes #19" appears to be exemplary. This work is representative of a re-discussion, through the reinterpretation of one of the NFTs that are part of the "Bored Apes Yacht Club" collection on Opensea, of both the artistic style (traced back to a primitive digital style that is demonstrative of its technological potential) and the belonging of the work to a platform it intends to distance itself from.

On Opensea, most of the collections that appear among the "Top" 50 “all-time” art category rankings, seem to demonstrate a degree of research (stylistic, technical, thematic) that is not comparable to that of the works that appear on the HEN platform.

The difference between the works that can be found on the two platforms seems to be similar to the one that typically separates commercial from cultural production; works designed for the market from works that do not satisfy the tastes and demand of the general public but that, guided by a spirit of experimentation, continuously elaborate new artistic practices, to be applied to what new NFT technologies allow. It is important to also point out the chromatic and stylistic research that characterizes all the works foregrounded by the two platforms: unlike in an exhibit, where the artists (or curators) decide and set the distance between the works and their position, the aesthetic impact plays a central role in capturing the viewer's attention. In art platforms, the visual placement structure offered to the viewer is predetermined, and yet the viewer can decide on which work to focus his attention.

Each work, therefore, must be able to attract attention to itself rather than to the other works besides which it is placed. Hence, together with the composition of the work, the choice of colour becomes a particularly

relevant element of first contact with the viewer who, after choosing it among many others, can then appreciate its content, details and features.

Among its top artworks, if compared to the "Top" collections in the Opensea “Art” category, HEN's most curated ones appear to reflect an artistic pursuit that focuses more on striking colour research. Examples of this are “School Band” by byhq and “oliv” by oshtru, as shown in Figure 7 and Figure 8.

Fig. 7

“School Band” by byhq, ranking in top 50 “hDAO” results

Note. From Hic et Nunc, by byhq, 2021, Hic et Nunc (https://hen.teztools.io/objkt/20476). Copyright 2021 by byhq

Fig. 8

"oliv" artwork by oshtru, ranking in top 50 "hDAO" results

Note. GIF. From Hic et Nunc, by oshtru, 2021, Hic et Nunc (https://hen.teztools.io/objkt/41476). Copyright 2021 by oshtru

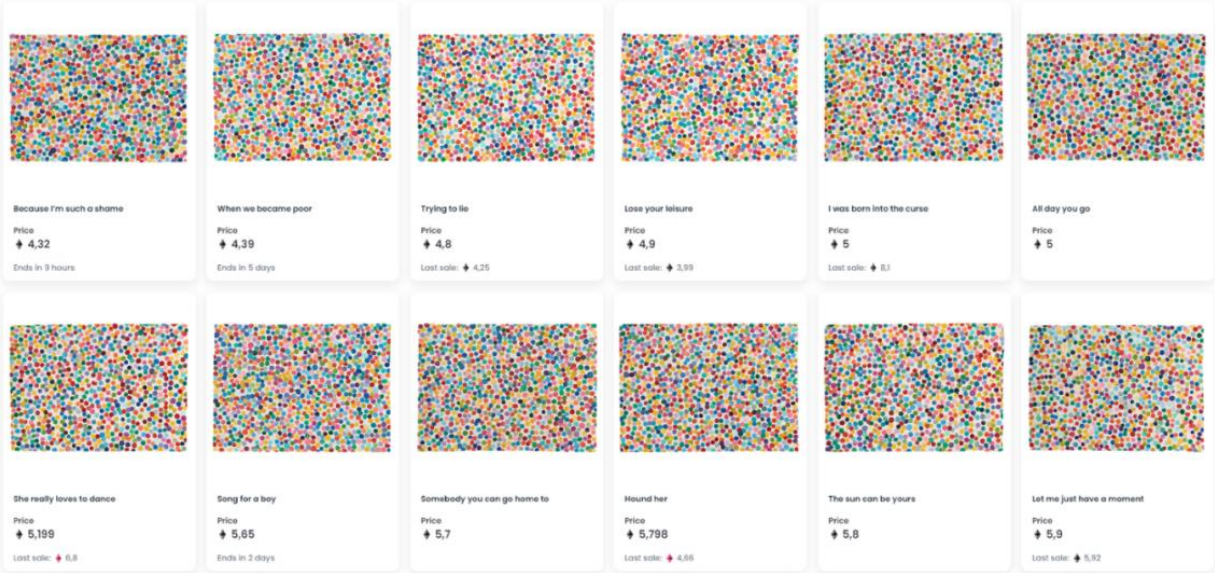
When the artist decides to produce generative artwork, as for the compositional elements, chromatism becomes of secondary importance, as the colour scheme will be determined later by algorithms. As mentioned above, HEN does not offer artists the feature of uploading collections as is the case for Opensea. Therefore, its more perceivable inclination towards individual artworks (which an artist could also have conceived as part of a series of works), if compared to the collections featured by Opensea (which above all tend to be generative art projects), seems to push toward an art based on stronger aesthetic research, free from any other utility constraint, which most of the "Top" ranked projects featured on Opensea tend to rely upon. Among Opensea's "Top" ranked generative artworks, there are examples of artworks that recall contemporary art, where fixed subjects and varying secondary elements of the composition arranged in the artworks through generative art techniques compose collections, as in "Cryptopunks", "Bored Apes Yacht Club" and "Lives of Asuna".

Opensea's more economically driven environment also attracts artists who have a certain reputation or are already established, inevitably invoking an artistic style that is closer, compared to a platform such as HEN, to the more traditional world of contemporary art. In particular, this is the case for the collections that rank in the top 20 of Damien Hirst's "The Currency", Tyler Hobbs' "Fidenza", and Dmitri Cherniack's "Ringers" by Art Blocks Curated as shown in Figure 9, Figure 10 and Figure 11 respectively, a platform which combines creative coding and blockchain technology to support artists in the launch of their projects.

Although these artists experiment using different digital-blockchain-related techniques in the creation of their works, they inevitably carry their culture with them, as can be seen by the style that seldom deviates from the artistic works they have previously produced.

Fig. 9

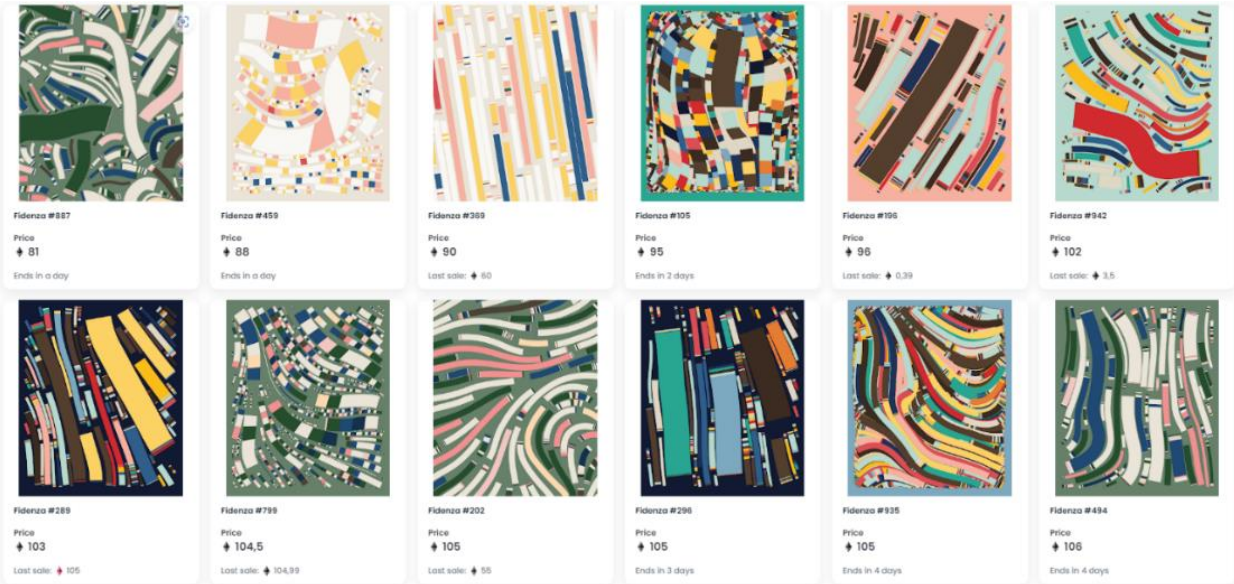
"The currency" collection by Damien Hirst, ranking in "Top" 50 "all-time" results of "Art" category



Note. A series of works part of the "The currency" collection on Opensea.

Fig. 10

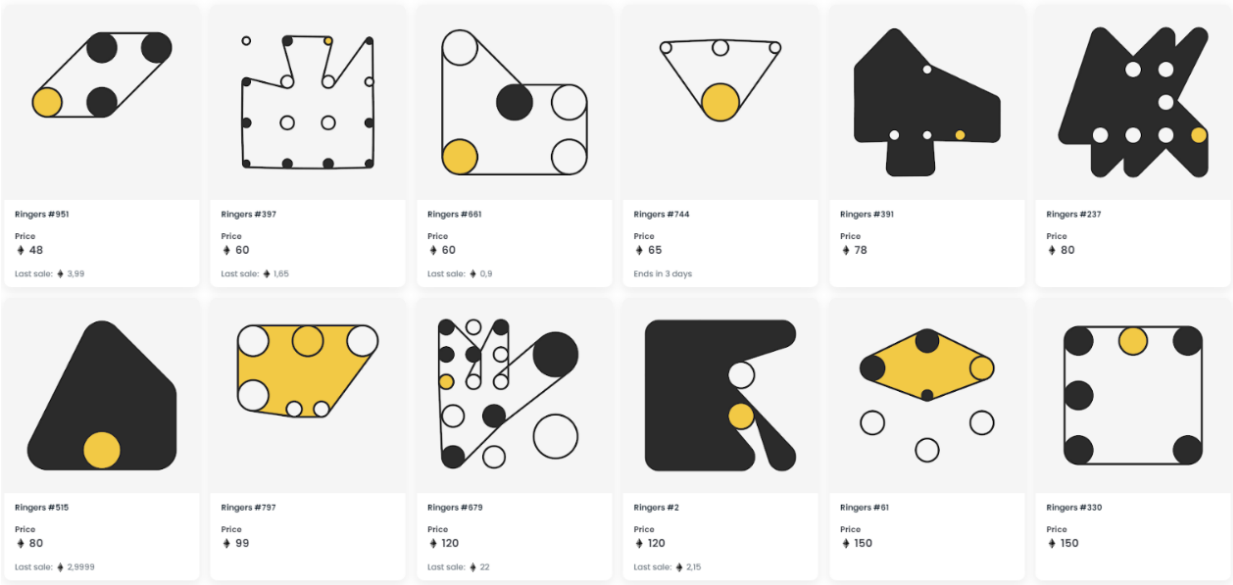
"Fidenza" collection by Tyler Hobbs, ranking in "Top" 50 "all-time" results of "Art" category



Note. A series of works part of the "Fidenza" collection on Opensea.

Fig. 11

"Ringers" collection by Dmitri Cherniack, ranking in "Top" 50 "all-time" results of "Art" category



Note. A series of works part of the "Ringers" collection on Opensea.

The descriptions of these artworks, seem to establish a game of messages with the public. Examples of this, as can be seen in Figure 9 is "Ringers" by Dmitri Cherniak", described as "ways to wrap a string around a set of pegs". In "Fidenza" by Tyler Hobbs, as shown in Figure 10 instead, "the varieties of scale, organization, texture, and colour usage it can employ create a wide array of generative possibilities". Lastly, Damien Hirst's "The Currency", as shown in Figure 11, "explores the boundaries of art and currency - when art changes and becomes a currency, and when currency becomes art". In such regards their belonging to the world of contemporary art seems to prevail, in line with Queiroz's interpretation of conceptual art "as if it was a private joke between artists, the wealth of reproduced and legitimated art, and the new public, knowledgeable of a pop expressiveness and a media culture that is increasingly dense" (Queiroz, 2017, p.1).

As is the case with the "Art Blocks Curated" works, certain works appear to be simply transported from the museum setting to that of virtual platforms, as if the platforms were merely a different environment into which artists can transpose their style. Certain works, on the other hand, are the result of an ad hoc production, conceived and produced for the virtual dimension to which they belong, but in which stylistic and compositional simplicity prevails, in comparison to the more varied artistic tendency in terms of styles and techniques that is used on artworks in HEN.

In both cases, however, we can find examples of generative art, where the artist has relied on the seemingly random production of the algorithm, even though the subject and the result of the generative production are always inevitably conditioned by the artist's initial approach.

4.2 Aesthetics of Artists' Presentation

NFT platforms serve as spaces for crypto artists and audiences to meet. In particular, NFT platforms allow users to create a personal profile where they can dispose of NFTs minted and collected in the platforms' supported blockchains. Through these platforms, users complete transactions that are recorded in the blockchain: each account is linked to a personal virtual wallet which, once connected to the platform, allows them to create a personal profile within it.

For both Opensea and HEN platforms, the users can create a profile that does not distinguish their role as collectors or artists. As shown in Figure 12 and Figure 13, Opensea includes a wider variety of sections (such as "Collected", "Created", "Favorited", "Activity", and "More", which includes "Offers received" and "Offers made") in order to filter through the artists' overall performance and activities in the platform, while Hic et Nunc's more minimal design is limited to the first two, placing "creations" before collected artworks ("collection"). It appears clear how HEN's more liminal design appears to favour the artist, as "a way to get and keep control of their artworks and reap related benefits" (Franceschet et al., 2021, p.404) rather than Opensea, where the user page is rather dedicated and fitted to the consumer of art.

Fig. 12

User profile page view on Opensea

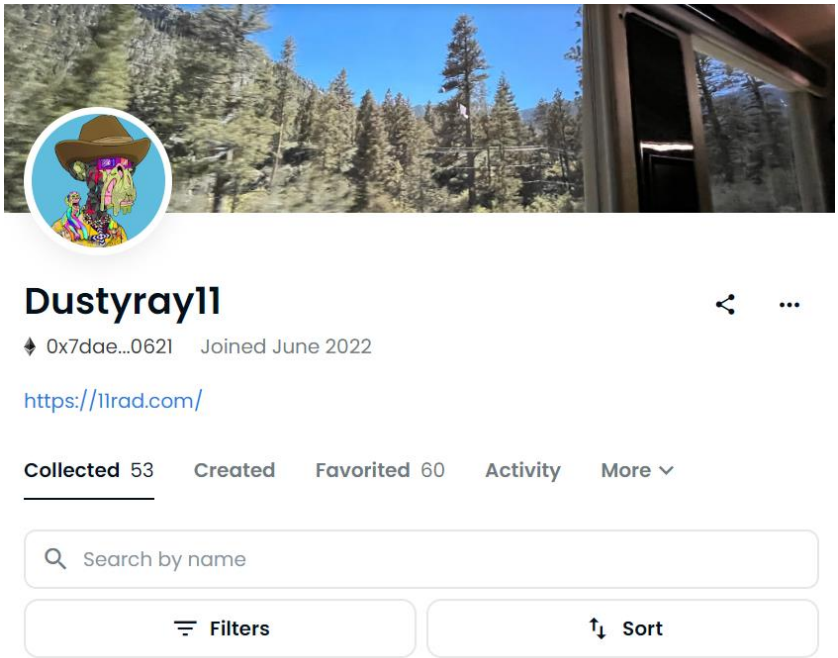
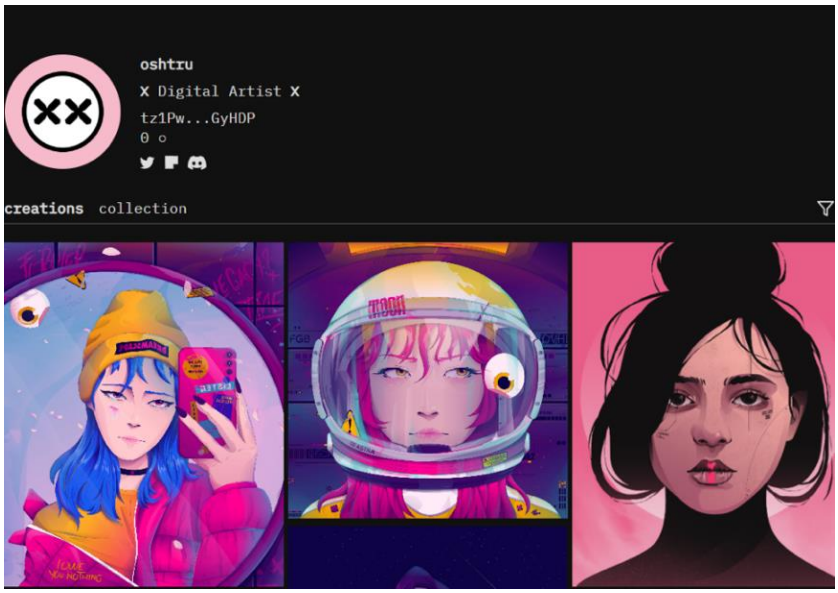


Fig. 13

User profile page view on Hic et Nunc



Unlike social media, neither platform offers any built-in communication feature, nor do they offer a space for users to expose their minted and collected works, however, both platforms include a feature that allows users to add links to other online spaces, which appear as icons in their profile pages. On Opensea users can include links to their Twitter profile, Instagram handle, or other hyperlinks (such as Discord, which appears frequently in users' profiles), on the other hand, HEN invites users to link their Twitter and Discord profiles (as well as GitHub and website pages) to a selected Tezos profile, a service by Tezos which allows users to associate public credentials with their Tezos account. Additionally, both platforms include a space for short descriptions where artists can add more information related to their works and activity.

It is evident that both of the platforms' communities reflect the values of anonymity, one of the underlying values of blockchain ideologies (Bsteh, 2021). Most of the top-ranked artists appearing in HEN's "hDAO" feed use this space to add additional information about themselves and their art. Exemplary of this are "Multidisciplinary Artists/From Traditional Painting, Photography to Musician" for the artists RainbowSprinkles, as shown in Figure 14, to more simple and less formal descriptions such as "love drawing" or " ponyuuu is a girl from ponyu universe that befriends anyone who also live in ponyu universe" for artist byhq and ponyuuu, as shown in Figure 15 and Figure 16 respectively.

Despite the wide variety of styles among the top voted "hDAO" artworks of artists on the HEN platform, in most cases anonymity remains at the center of artists' presentation, with artists coming up with creative names that have no direct reference to their real identity.

Fig. 14

Artist RainbowSprinkles, personal profile page on Hic et Nunc

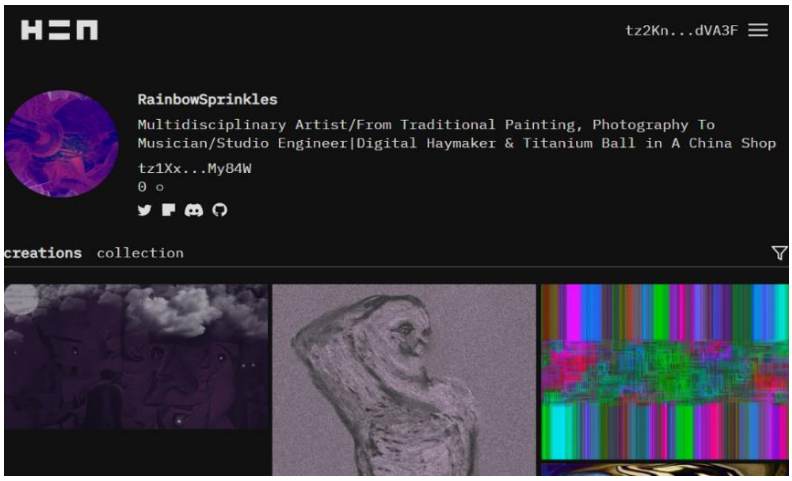


Fig. 15

Artist byhq, personal profile page on Hic et Nunc

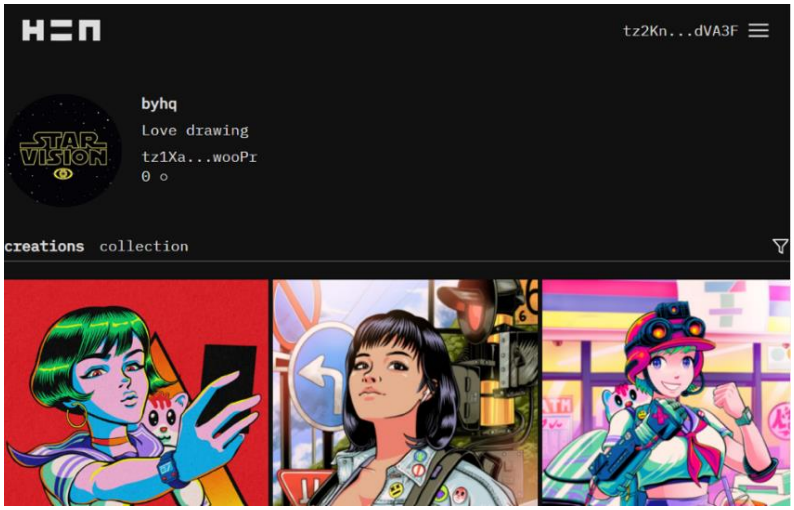
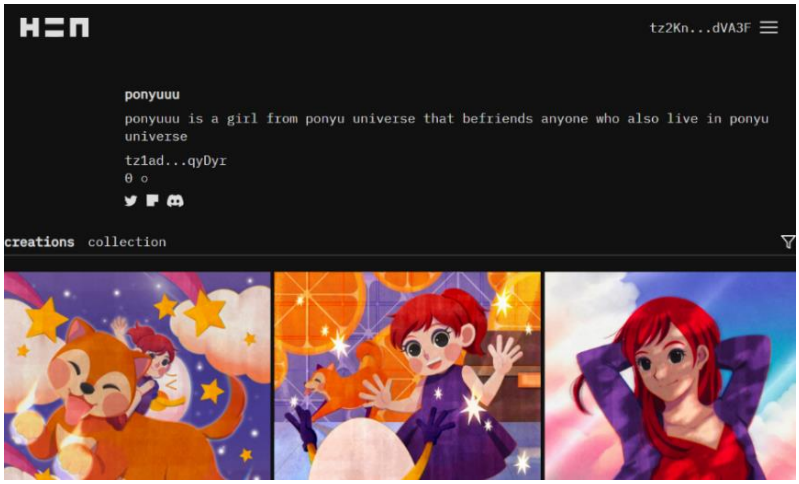


Fig. 16

Artist ponyuuu, personal profile page on Hic et Nunc



On the other hand, among the authors of Opensea's "Top" art collections, artists' identities seem to assume a secondary role as they are rarely mentioned in most of the respective collection pages (as is the case for "CryptoPunks" and "Bored Ape Yacht Club" collections), and in the rare cases they are not omitted, they are not published by the artists themselves directly (not linked to the artist's profile of that collection within the platform). In Opensea, among the collections categorized as "Art" that occupy the top 50 positions by sales volume, anonymity does not appear to reflect the same values it has on HEN; it rather reflects a secondary importance of the artist, left behind the utilitarian value of ownership of the assets and the commercial aims of NFT projects of which artists are just part of.

However, among the "Top" results on Opensea's "Art" category, some artists present their artworks using their real identity instead of aliases. This appears to be the case especially for the popular collections of already acclaimed contemporary artists, such as Damien Hirst with "The Currency" collection, and the Tyler Hobbs and Dmitri Cherniak collections already mentioned in the section above. As mentioned in the previous section, these artists do not directly present themselves on the platform and rely on an external service (that of "Art Blocks Curated") which takes care of the presentation of their artworks within the platform of Opensea.

Fig. 17

Collection page of "The currency" by Damien Hirst, ranking in "Top" 50 "all-time" results of "Art" category on Opensea

Fig. 18

Collection page of "Fidenza" by Tyler Hobbs, ranking in "Top" 50 "all-time" results of "Art" category on Opensea

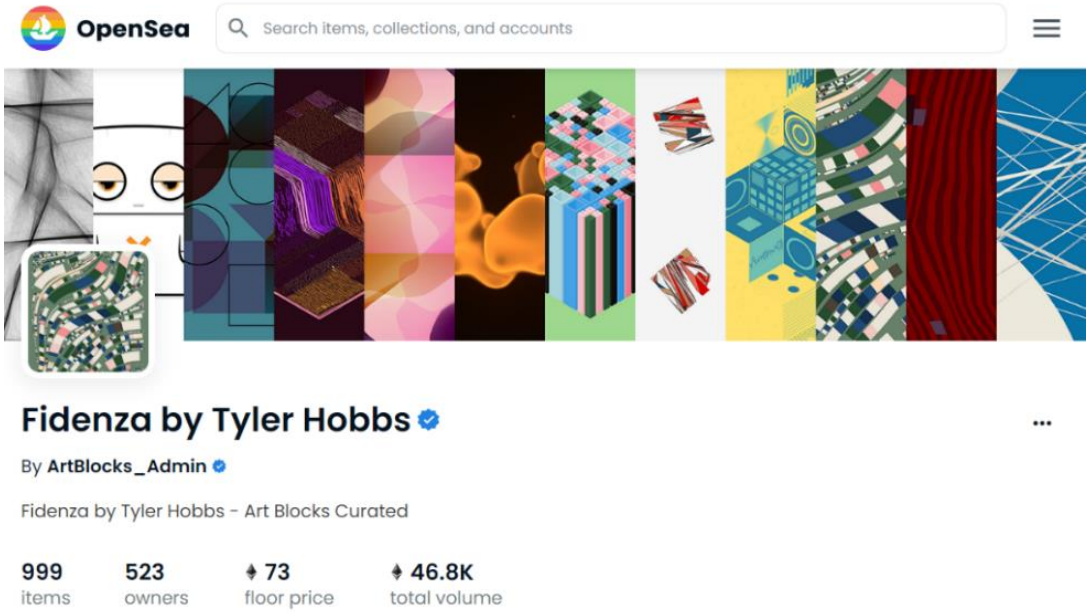
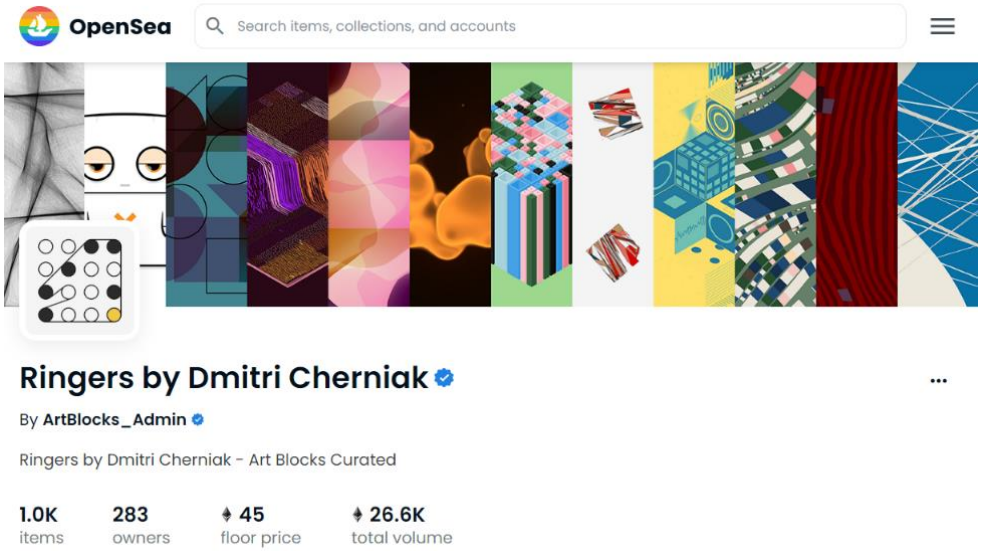


Fig. 19

Collection page of "Ringers" by Dmitri Cherniak, ranking in "Top" 50 "all-time" results of "Art" category on Opensea



As can be observed from the top 50 collections that Opensea categorizes as "Art" through its search filters in most of the cases artists identities are not directly included: in most cases, these works appear to be part of larger collectives or groups of which the artists are just a part of.

Despite all that has been said above, HEN is overall largely un-curated. Many artists — even well-known artists — avoid using their names and are known only by their online aliases. As anonymity highlights the

distancing of artists from the legitimacy frameworks of what could be considered the traditional art world, this value appears in contradiction with the ranking systems that guide users' navigation on Opensea.
Obviously, the identity artists choose is maintained across all the other social media platforms they redirect users to through their personal profiles on HEN to interact with their audience. In fact, the promotion of artworks and, more generally, the artists' self-branding practices do not take place directly in the platforms. Artists do, instead, tend to link their Twitter accounts and other social media handles where such activities take place. In such terms, artists' presentation on HEN seems to differ from that of Opensea. Among the "Top" 50 ranking artworks in the "Art" category, the value of anonymity of artists on Opensea (which as argued in most cases are not even included) seems to be leveraged as a means to focus on the artwork as a utilitarian product, rather than represent the belief of art's liberation through the visions and ideologies underlying the NFT phenomena. The concept of commodification of art represented by Opensea's volume ranking-based navigation model hence appears to strongly reflect on the ways in which artists' promotion manifests itself.

Among the top-ranking collections on Opensea that explicitly include the artist's identity (such as those curated by "ArtBlocks", which include the previously discussed artworks by Damien Hirst, Tyler Hobbs and Dmitri Cherniak), the artist's social media accounts are not provided in the collections' page in the platform itself. In contrary to Opensea, in HEN, most artists who independently manage their personal profiles, directly link their social accounts or personal website pages on the platform. Observing artists' Twitter profile pages, it appears as most top-ranking artists in the "hDAO" feed of the HEN platform mainly focus on the visual aesthetics of their profiles, almost as they were in competition for the most appealing graphics and presentation, in line with the crypto and digital art aesthetics. As argued by Kang et. al (2019) in relation to enhancing kinestetics and emotional response of audiences through interaction in social media, also their presence on these social platforms can help promote their artistic activity and position as artists in the competitive NFT market.
The active engagement in visual elements and selected profile images finds its reason in the artists' necessity for self-legitimation, and, to that end, in their need to reinforce their autonomization though entrepreneurial activity. As can be seen in Figures 20, 21 and 22 considering their engagement as self-employed entrepreneurs, as suggested by Kackovic & Wijneberg (2020), in promoting their artistic activity across social media from the identities portrayed through images, profile descriptions and language used, artists tend to display traits of innovative, less conditioned and more singular artistic orientation.

Fig. 20

Twitter profile page of artist RainbowSprinkles (artist from Hic et Nunc)

Fig. 21

Twitter profile page of artist ponyuuu (artist from Hic et Nunc)

Fig. 22

Twitter profile page of artist RainbowSprinkles (artist from Hic et Nunc)

As for the Twitter profiles of artists already legitimized outside of the context of the NFT space (appearing in Opensea's "Top" 50 all-time "Art" rankings), their presentation seems to represent a more formal and simplified online identity, closer to what could be understood as a professional or formal presentation. Examples of this are shown in Figure 23, Figure 24 and Figure 25, for the Twitter profiles of Damien Hirst, Tyler Hobbs and Dmitri Cherniak respectively.

Fig. 23

Twitter profile page of artist Damien Hirst (Opensea)

Fig. 24

Twitter profile page of artist Tyler Hobbs (Opensea)

Fig. 25

Twitter profile page of artist Dmitri Cherniak (Opensea)

Although HEN's platform's design seems to suggest a more minimal and less commercially oriented visual aesthetic if compared to the artists of collections curated by "Art Blocks Curated" in Opensea, artists' presentations in the latter appear more simplistic because their positioning is determined by already established dominant voices in the field.

This difference is also visible for both platforms in the description sections of the artists' Twitter profiles, which tend to present themselves differently.

As shown in Figure 21 and Figure 22 respectively, descriptions such as “I create weird and kind Creatures!” by @edenftCreatures or “We're Mr. & Mrs. Ponyuuu!” for @ponyuuuNFT suggest a language that is in contrast with that used by acclaimed artists of Opensea. While in HEN artists’ present themselves in unconventional ways compared to the above-mentioned more formal style of language used by already legitimized artists in contemporary art on Opensea’s platform, on Opensea, as argued in regards to the aesthetics of their profiles in the selection of profile images, artists tend to limit themselves to more simple and functional descriptions. On the other hand, the presentation of artists of independent collections of Opensea appears to be very similar to that of artists of HEN.

As most of the times, these artists are not mentioned directly within the platform’s collections, researching for Twitter profiles of artists of the “Top” ranking independent “Art” collections on Opensea who are not mediated or curated by external sources, it appears that they tend to explicitly reference their involvement in projects that have already acquired legitimation in the NFT space. On the contrary in HEN acquired legitimacy is not seen as a core value and unpopularity appears to be more valued. Examples of this are shown in Figure 26, Figure 27 and Figure 28 which present the Twitter accounts _migwashere, hagglebub and 3landboy (artists of the collections of “Bored Ape Yacht Club”, “Lives of Asuna” and “3Landers” respectively) ranking among the top 50 collections by volume of sales on Opensea.

Fig. 26

Twitter profile page of _migwashere*, one of the artists of “Bored Ape Yatch Club” (Opensea)*

Fig. 27

Twitter profile page of artist hagglefish, artist of "Lives of Asuna" (Opensea)

Fig. 28

Twitter profile page of artist 3landboy, artist of "3Landers" (Opensea)

Although artists are not directly mentioned or promoted in most independent collections of Opensea, there appears to be a correlation between the ranking of collections in the Opensea "Art" category and the number of followers the respective artists count on social media. In a commodified digital art market environment such as that of Opensea, the number of followers can be understood as a means for the collectors to evaluate an artist's legitimacy, as an indicator of social status, and so ultimately accounted for when evaluating the

potential profitability that the investment could bring to its buyer. In this consideration, for the platform of Opensea, the social attention a digital object attracts appears to be more valuable compared to the actual message of the content transmitted (Mitnick,2010), in this case in comparison to the art object presented. In HEN instead, the status of an artist on social media appears to be secondary to the aesthetics of its art and this factor does not appear to find such a close correlation for the artists' ranking in the "hDAO" feed, as much as it does for Opensea.

Although arguably Opensea's general higher artists' follower counts can be explained by the platform's popularity within the NFT field, it remains a significant element, considering that in such a volume and sales-driven platform, follower count is determinant in users' investment decisions and, in order to improve self-branding, artists might engage in different strategies to acquire more followers and increase their status.

4.3 Aesthetics of Governance of Hic et Nunc and Opensea

Opensea is a private company that, as stated in the platform's terms of service "provides peer-to-peer Web3 services that help users discover and interact directly with NFTs on the blockchain". HEN instead, as defined in the platform's "About" page, presents itself as a "decentralized application [which] allows its users to manage decentralized digital assets, serving as a public smart contract infrastructure on Tezos Blockchain".

Hicathon's first press release (posted on the dedicated website page for the first hackathon organized by the founder which took place between May 22nd and May 23rd 2021, as an occasion for the community to participate in the platform's development) describes the vision of its founder Rafael Lima, as "experimenting with creative economies as an alternative, also to the traditional DeFi (decentralized finance) mode of thinking" (aljaparis, 2021, para. 4).

This statement clearly expresses HEN's intent to position itself in contrast with its competitors in the NFT space, where most platforms, such as Opensea, appear to actively engage in mediating user activity through untransparent centralized frameworks of governance and data management. Rooted in the vision for decentralization, the platform was developed with the community's participation in what could be defined as an initial framework of distributed governance (Vertical Crypto Art, 2021).

Although the platform cannot be defined as a DAO, as it was ultimately centralized, and fully controlled by Rafael Lima, the founder's vision and engagement with the community was rooted in a model oriented towards achieving a full distributed governance, intended as "the means of achieving the direction, control, and coordination of stakeholders within the context of a given blockchain project to which they jointly contribute" (Pelt et. al, 2021, p. 27).

Lima managed to successfully run the platform single-handedly and with the help of contributing developers. The major contribution of the community that characterized the development of HEN, only two weeks after

its launch, on March 1st 2021, was when Rafael Lima organized a hackathon. As shown in Figure 29, for this event Lima created a dedicated website for the community to keep updated and follow the developments of the platform. In this hackathon, named "Hicathon" by its founder, 150 users from 77 different countries participated. In the "about" section of HEN's dedicated Hicathon page, this event is described as an opportunity for participants to "work on accelerating the development of H=N by working alongside the devs & governance" (homeworkpunks, 2021). Later, in September 2021, 7 teams met again to transform these prototypes into products ready to be implemented on the platform.

Fig. 29

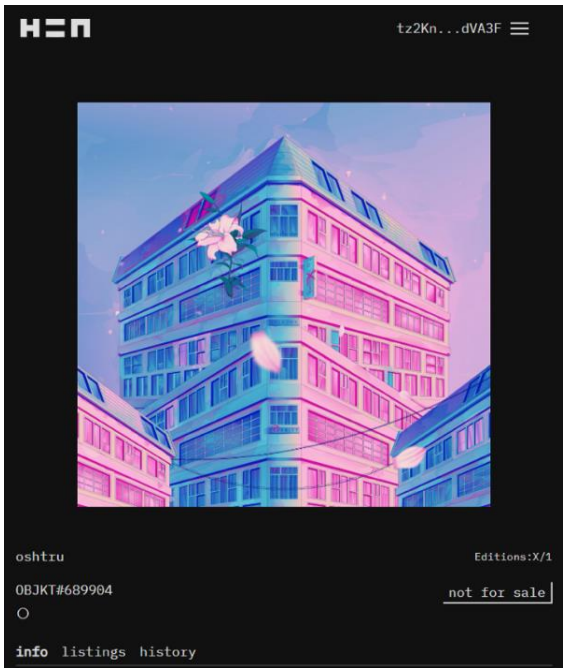
Hicathon dedicated page of Hic et Nunc

Among the many accomplishments of this event, for HEN, one of the most significant implementations was the hDAO token, which was intended to be the governance token for the Hic Et Nunc platform.

With the issuing of this token, a participatory system of ranking was introduced and set as a filter to navigate artworks on the website's landing page. As shown in Figure 30, this governance model, based on the hDAO token, allowed users to upvote NFTs minted in the platform by simply clicking on the "curate" function that appears as a circle icon placed on the bottom left corner ("curate" function), below the preview of artworks. In this way, the most "curated" artworks would then appear at the top of the search filter's feed.

Fig. 30

HEN's artwork preview page



During the first 45 days after Hic Et Nunc went live, every sale and purchase of an NFT gave users free hDAO tokens. Once the initial distribution of these tokens (which could be acquired by simply conducting transactions in the platform) reached the capped supply of 650000, hDAO tokens were included in the ecosystem of Quipuswamp (a crypto online exchange) where they could be traded with other cryptocurrencies in the Tezos blockchain.

Taking into consideration the ways in which the two platforms mediate the presentation of artworks, it is evident that their two different navigational models are strongly determined by their statute, which is centralized in both cases but based on very different visions and governance structures on which the platforms are designed.

In the case of Opensea's corporate commercially oriented platform, the perspectives by Langlois and Elmer (2013) as well as Plantin et al. (2016) in regards to how economic interests of corporate platforms affect the interface designs appear to resonate more than ever. The presence of "Top" results based on the volume of sales which are repurposed throughout the platform in multiple sections, can be interpreted as an attempt to ease the search for the most economically valuable NFTs and by directly promoting them in this way, increase its earnings through the 2.5 % commissions taken by the platform on every transaction.

On the other hand, it appears clear how, although also requiring an equal commission per transaction used for the maintenance of the platform, the different vision for HEN does not appear to favour such economic interests. In such regards, the "hDAO" homepage filter feed determines a ranking based on an appreciation of art, and the filters through which to view the most sold artworks across different time periods are placed at the end of this list of the platform's filters.

Along the foregrounded “Top” ranking artworks presented in the “Stats” section, Opensea also allows users to navigate through the most sold artworks, according to the volume of sales, directly from its Homepage. The platform also seems to manifest its active role as a mediator in its less objective and more questionable navigation sections and filter categories. As shown in Figure 31, this is the case for the default “Trending” filter in the “Explore collections” section (also repurposed as a secondary filter in the “Stats” section following the “Top” category) as well as the “Notable Drops” section present in the platform’s homepage, as well as the daily picked artworks appearing in the platforms’ landing page, as can be seen in Figure 32.

Fig. 31

Opensea’s “Explore collections” section

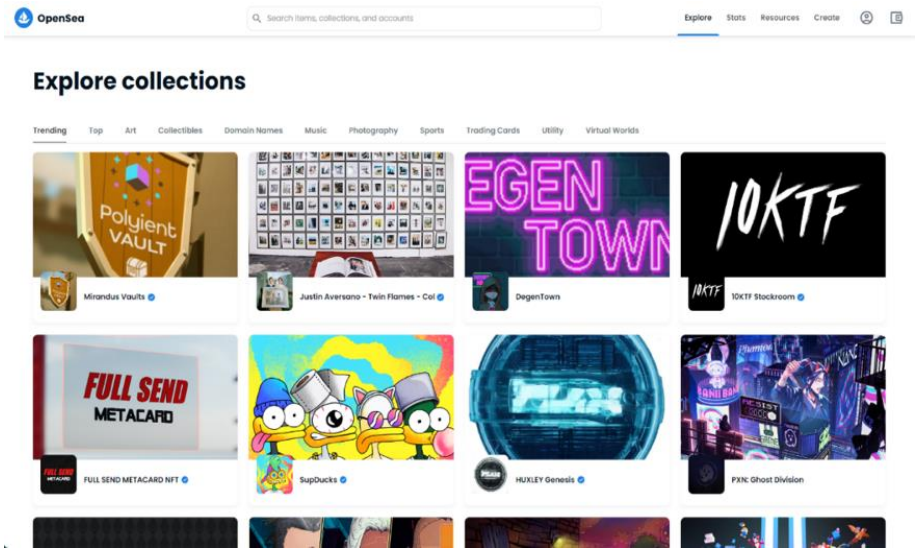
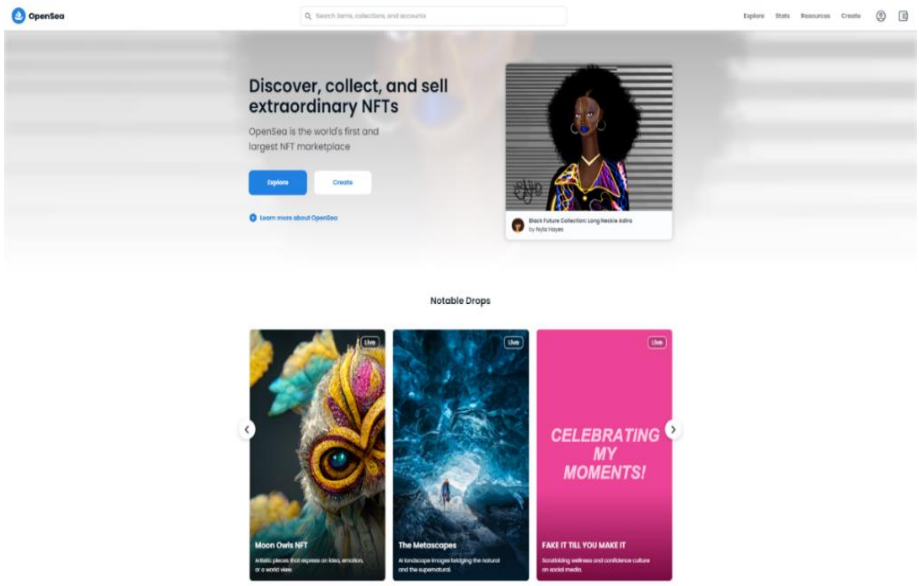


Fig. 32

Opensea’s landing page



For both the “Trending” default filter in the “Explore” section under the platform’s top navigation bar, as well as the homepage’s “Notable drops” section, the mechanisms through which artists are selected to be featured in these sections are unclear and the processes and criteria of selection of such artworks are unknown. As shown in Figure 33, the only information provided by Opensea in regard to the processes of selection of artworks to be featured on its homepage was given in a tweet published on the 10th of July 2021, where it invited users to share their minted artworks by posting Opensea’s NFT link, tagging the Opensea official Twitter page and adding “#OpenSeaNFT”.

Fig. 33

Tweet post by Opensea describing how to get featured on the platform’s homepage

Besides this tweet, Opensea never provided any further clarification on which criteria are used to select artworks to be considered for being featured on the platform, nor regarding who is in charge of making such decisions, leaving it unclear whether such responsibility is assigned to an external specialized team of critics, experts or art curators, an internal dedicated team, or a single employee within the corporation.

Opensea’s role in actively promoting some artists instead of others also appears evident in regards to its verification feature. The verified checkmark badge attributed to a collection not only arguably ensures more safety to users in regards to the authenticity of an artist or a collection minted on the platform, but it also inevitably affects the audience’s perception, as verified works have greater appeal than non-verified ones. As opposed to the non-disclosed criteria the platform adopts to promote and feature certain artists’ works instead of others, the conditions for verification of an account or a specific collection are stated in the platform’s “Help Center” page under the “User Safety” section, where it is stated that “Accounts that own collections with at least 75 ETH of volume sold (or equivalent in SOL) and meet other criteria like minimum activity levels and social presence are eligible to apply for verification”. In this case, the decision to validate accounts appears to

be based primarily on the economic aspect of the sales volume, reconfirming Opensea's economic inclination to consider art's value in relation to economic capital: an expression of art's commodification to its fullest.

As for HEN, it disposes of a verification badge to authenticate users, but, unlike Opensea, there are no specific requirements to meet and any user can request to be verified. In fact, the process for verifying one's account in HEN simply requires one to send a small amount transaction to HEN's specific verification address and to fill out a google form confirming ownership of the account through a required Twitter profile. Because this process does not appear to serve the same purpose pursued by Opensea's verification models, it mainly acts as an ulterior measure that helps users identify copyminters or stolen identities. Still, HEN invites users to adopt measures of self-caution. In the "about" section of the platform, it states: "We're concerned about your security and autonomy. please verify the seller's information while making transactions". Nevertheless, Opensea's verification feature, in addition to its above-mentioned role in the selection of artworks to feature in the platform's navigation sections, appears to contradict the terms of service that state that: "you bear full responsibility for verifying the identity, legitimacy, and authenticity of NFTs that you purchase from third-party sellers using the Service and we make no claims about the identity, legitimacy, functionality, or authenticity of users or NFTs (and any content associated with such NFTs) visible on the Service."

While both Opensea and HEN seem to flaunt a belonging to the values of decentralization and autonomy of the artist, the curatory and moderatory practices adopted by the platforms according to their respective governance models, suggest two different conceptions and approaches. Although Opensea claims to not exercise any type of control over the NFTs within the platform, being a corporate platform with dominant commercial interests, it inevitably adopts curatory, moderatory and regulatory practices that develop in parallel with the platform's frameworks of corporate responsibility. In its terms of service, Opensea states: "We do not have custody or control over the NFTs or blockchains you are interacting with and we do not execute or effectuate purchases, transfers, or sales of NFTs […] OpenSea may require you to provide additional information and documents in certain circumstances, such as at the request of any government authority, as any applicable law or regulation dictates, or to investigate a potential violation of these Terms. In such cases, OpenSea, in its sole discretion, may disable your Account and block your ability to access the Service until such additional information and documents are processed by OpenSea. If you do not provide complete and accurate information in response to such a request, OpenSea may refuse to restore your access to the Service". Gillespie's (2017) claims about the tensions between platforms' claims to impartiality and their actual manifestation as intermediaries, appear relevant in Opensea's case, as its mediation manifests in proprietary and opaque ways, making platforms arbiters of cultural value. On one hand, Opensea affirms it has no control over users' on-chain NFTs and, on the other, it appears clear how user accounts are under its control, given the platform's ability to disable or freeze accounts at its sole discretion.
Opensea exercised such discretion when, following the Russian invasion of Ukraine, Western countries required international companies to take sides by participating in sanctions imposed against Russia by the

European and American governments. Like many financial institutions and multinationals, Opensea decided to join in, blocking the operation of certain categories of platform users, chosen for their nationality, who were prevented from carrying out any activity on the trading platform. Users from Iran, whose protests became public, complained that they could not sell or buy NFTs and stated that collections, purchased before Opensea's adherence to the international sanctions against Russia and countries that supported Russia's invasion of Ukraine, disappeared from their profiles and turned into "404" pages.

Another event during which Opensea has directly engaged in similar activities, occurred on December 30th, 2021, when Todd Kramer, owner of New York's "Ross + Kramer Art Gallery", was the victim of a hack in his wallet through which 16 NFTs, valued at the time for a total of approximately 2.28 million dollars or 615 ETH (Jamie Redman, 2021). Although the platform itself bore no direct responsibility in relation to the event, OpenSea instantly reacted by freezing trades on the stolen NFTs. This action was not only heavily criticized by Opensea's community but also led many to further question OpenSea's claims to decentralization.
Although strongly based on ideologies and values of decentralization, Rafael Lima recognized the necessity of moderatory and curatory practices. However, these practices are not as structured as in Opensea's corporate frameworks, rather these decisions are taken in response to single users' requests through the official Discord channel of the platform. Additionally, the criteria according to which such decisions are taken are clearly stated in the platform's terms of service, as opposed to Opensea. Examples of these criteria are "Copyminting" (understood as minting another artist's work without the artist's consent), "Wash Trading" (defined as manipulating prices by trading an own NFT through a network of addresses), "Minting identical NFTs on other platforms", and "Explicit, sensitive, offensive or hate discourses which doesn't present artistic intents" (as stated in the platform's terms of service accessible through HEN's GitHub page) (crzypatchwork, 2021).

On the matter, in HEN's dedicated community forum, in a thread named "Restricted OBJKTs", presenting himself through the alias "crzypatchwork", Lima suggests: "Reports can be made through our discord channel in the #report-copymint section, through e-mails or posting an issue in the following repository hicetnunc-reports 22. Contesting those restrictions can be made through these very same channels […] Due the fact that hicetnunc.xyz is a decentralized application, restricted OBJKTs are only superficially restricted meaning that they are still available through our APIs, and through on-chain data. Tools which allows accessing them without restriction nonetheless should be further developed to guarantee the liberty of users and for them to not be dependent exclusively on hicetnunc.xyz" (crzypatchwork, 2021). Following the thread Lima also suggests that more decentralized solutions, where such decisions are taken care of by the community, could be developed during the next phases, stating: "A decentralized solution would better allow us though to make such process transparent, which would be an interesting contribution to the web3 space the development of a tool for reports/contesting, as a trial on-chain."

Although HEN's community did not accomplish such achievements before the platform was shut down by its creator, its tendency to detach from its role as a mediator between artists and users is also confirmed by the NFT storage mechanism, on which the platform is designed. Developed by the Hicathon participants, HEN's NFT smart contracts allow for the minted artworks to be stored externally, as IPFS uses a decentralized peer-to-peer network to host web pages on the internet. In HEN's website "about" section, HEN states "IPFS OBJKTs can be minted and traded by permissionless means. such experiment was designed intending to imagine alternative crypto economies".
Using IPFS file storage mechanisms guarantees the permanence of the artworks' data to be independent from a single centralized server, which was originally the case for the Opensea platform and still appears to be the case for some popular NFT marketplaces. This allowed for the portability of data across other services and for the community to retrieve the metadata linked to the OBJKTs in HEN even after the platform was suddenly shut down. In fact, IPFS being an open API allows to easily transfer data from one IPFS storage node to another one, as it allowed to query current host nodes and import all the NFT assets automatically, without having to manually download and re-upload each of the files outside of the IPFS network.

As Opensea was initially designed to store the NFTs minted in their platform on their centralized servers, only later in June 2021, the platform added the feature to store NFTs with IPFS and Filecoin (a blockchain-based service that allows users to rent out their digital storage space) launching the feature that allows to freeze NFT metadata. Until then, if Opensea's servers went down, the NFT assets which were not stored directly on the blockchain but were on the platforms' centralized servers, would become tokens not pointing to any media files, resulting in a loss of the assets' metadata which the smart contracts linked to.

As Opensea must abide by government policies in its own corporate interest as a company registered and headquartered in the US, the platform's ability to remove NFTs stored through the implementation of IPFS is not fully overcome, but limited to the retrieving of the metadata in the platform. The implementation of this feature allows the platform to reconcile its moderator and regulatory necessities with the underlying visions of decentralization of the crypto community, arguably allowing for the platform to reach a higher degree of decentralization.

5. Conclusion

This thesis aimed to interpret from an aesthetic point of view the two social contexts of Opensea and Hic et Nunc, in order to highlight, in consideration of the ways they mutually affect each other (users-platforms) the legitimation systems the two popular NFT platforms promote.

The two platforms examined declare their structural differences starting from the navigation method they offer, that is, from the first contact the user has with the platform. The two different methods of navigating the platforms, on which the objects and subjects analyzed for each platform in the research were selected, immediately determine hierarchies in the presentation of artworks that inevitably legitimize some artists more than others. Opensea promoting “Top” NFT collections — where artworks are ranked in terms of sales over different time periods — in numerous sections of the platform compared to HEN, appears to highlight a reinforced conception of art as a commodity, reflecting a mercantile culture, where the artist’s freedom is conditioned by the new art market.

This limited freedom also appears to be reflected in the same aesthetics of the artworks shown on Opensea, which in most cases seem much less diverse than they are on the HEN platform. Whether it is collections such as "Bored Apes Yacht Club", "Cryptopunks" or “Lives of Asuna" (the value of which is not determined solely by the works themselves but is based on the promises to will-be owners to participate in a project which the specific artwork is origin or part of), or whether it is, and this is more obvious, collections of artists already legitimized elsewhere in the art world (as in the case of the collections signed by Damien Hirst, Tyler Hobbs and Dmitri Cherniak), composition, techniques used and styles tend to resemble each other, as if to seek (or to produce) a standard taste. In HEN, among the artworks that are positioned in the top results of the "hDAO" filter on the homepage, the diversity of artistic expression is immediately evident, representing (but also making users perceive) greater expressive freedom granted to artists. The diversity and characteristics of the works in the "hDAO" decentralized voting system for sorting artworks in the platform’s homepage default search filter suggests how this digital environment requires artists to carry out continuous artistic creative innovation by promoting greater competition among them. This manifests in the experimental and creative aesthetics of artworks of HEN, rather than what could be defined as a more decorative art style such as that observed in the “Top” ranking “Art” collections on Opensea.

Faced with certain examples of contemporary art and, even more so, of crypto art utility-based projects, one often wonders where the boundaries lie between creative freedom and the deliberate construction of products selected on the basis of the public’s tastes or on their saleability. The similar styles and formats of generative art collections which dominate Opensea’s rankings, where the artist-author’s name often is not even mentioned, seem to suggest that, in certain contexts, it is the latter that prevails.

While Opensea’s “Top” ranking collections are products of artists who are usually part of collectives (joining a collective naturally limits the independence of the individual artist), the increased engagement in innovative

artistic activity in terms of aesthetics of artworks of HEN, can be understood along the lines of self-employed artists, as suggested by Kackovic and Wijneberg (2020).

The more capitalist and commodified dimension of Opensea is confirmed by the structure of the users' profiles determined by the platform itself which, places the "Collected" section before the "Created" NFTs. On the contrary, by placing "creations" before "collections", HEN seems to be more respectful of and dedicated to the artist's needs: user profiles on the platform of HEN almost become personal galleries where the audience is directly presented with all of the artworks created by a specific artist. Overall Opensea appears to be a platform tailored for the needs of collectors rather than artists. This also appears to explain, along with the secondary role that artists assume in most "Top" ranking "Art" collections, the fewer presence of artists recurring to the use of personal profile pages to present and promote their artistic activity compared to HEN.

As for the presentation of artists, in both cases, most artists strengthen and support their activity through social channels. This parallel presence plays an important role in determining (and, for the purposes of this study to interpret) the legitimation patterns of the two social contexts.
With the exception of artists legitimized by external third-party agencies and services, such as "Art Blocks Cuarted" (which represent a minority compared to the very numerous independent NFT collections that dominate its "Top" art rankings), the vast majority of artists ranking as "Top" collections in Opensea's "Art" category seem to follow the well-established trend of choosing anonymity, as is also the case for artists in the platform of HEN. For both platforms, artists present themselves through aliases that they maintain across social platforms, as seen in this thesis for the case of Twitter. Without delving into the theme and its complex reasons, it is important to highlight the coherence of this trend with the founding principles of this art: blockchain is anonymous (Bsteh, 2021), crypto art provides for anonymous transactions, and the platforms are mostly inhabited by artists who prefer to stay anonymous or to use a pseudonym instead of their own name.
The author's anonymity makes the work less recognizable, presented as an orphan of its creator. The author renounces to be indissolubly linked to his work, unlike what in traditional art constitutes one of the classic principles of legitimation: the explicit and indisputable interdependence between work and author. The impersonality of the artist (especially when he presents himself under a false name or through an alias) removes any possible racial or gender prejudice, determining the creation, around the work and the genre, of a community that is inclusive and differentiated by statute.
In certain instances, as is the case for both Opensea and HEN, anonymity appears to be a choice, but as far as Opensea is concerned (excluding the collections of particularly well-known artists, such as Damien Hirst, Tyler Hobbs, and Dmitri Cherniak, curated by external sources), most of the "Top" ranking art collections often do not appear to base their value on the artist: most of the collections, in fact, do not to indicate the author. In these collections (which as argued appear to be mainly crypto art projects linked to a utility derived from the ownership of the asset), the absence of artists' aliases or social media links in the collection pages, rather than representing a reaffirmation along the lines of the values of anonymity that is typical of the inspiring

technology, it more directly simply responds to the commodification and profit maximization objectives that crypto art seems to have chosen as guidelines at this stage, objectives that in the case of Opensea make authorship almost irrelevant.
Still, the artist does not always and completely remain on the sidelines: most artists present themselves on social platforms, where they can obtain significant results in terms of followers.
Unlike Opensea, it is precisely the artist-artwork duo that dominates the top "hDAO" results in HEN, though mitigated by the fact that artists seem to value unpopularity.
Although artists are involved in much more complex legitimation processes, the evaluation and validation role, exercised by the audience with the ease and "freedom of incompetence" that are typical of the web, can benefit certain artists instead of others, affecting their legitimation.

For the artist — already engaged in the progressive autonomization process to which the Web, and in particular Web 2.0, had already offered many tools and platforms — adhering to the logic, dynamics and rules of the social presence is not only useful but becomes necessary in order to be competitive in these digital art markets. This is evident, for example, by observing the profiles that the artists in both of the platforms examined have created on social channels, particularly, on Twitter (the channel preferred by those who work in crypto art). Once again, among the similarity of presentation in light of anonymity, the two platforms' different approaches are reflected in the behaviors of the two different groups of artists who inhabit them: the artists present on Opensea are more engaged in acquiring followers on social channels, often also legitimizing themselves with the publication of their digital curriculum vitae, reaffirming their participation to previous successful NFT art projects; on the other hand, those artists who are present on HEN, when they are active on social channels, seem to be more interested in reaffirming their identity and cultural belonging, where unpopularity rather appears to be more valued. It is precisely the artists present on Opensea who appear to resort more to self-promotion activities on social channels, also referring to the artworks they have participated to in the past, in order to increase their online visibility. The major engagement of artists of Opensea on Twitter can be understood in light of the secondary role artists assume on a platform dominated by NFT projects that are classified as "Art", but whose value does not appear to be determined by the author, aware of referring to a community that assigns importance to social status as an indicator of profitability. In this regard, as well as in consideration of the prices that NFTs of collections such as "CryptoPunks", "Bored Ape Yacht Club" or "Lives of Asuna" have reached, it is precisely Opensea's more commercially oriented dimension that seems to reflect the link between the value of an asset and its social status, as defined by Howcroft (2022) in relation to the volatility of the NFT market.
This activity is not to be understood exclusively as a response to the artists' search for greater autonomy, nor necessarily to their need to legitimize themselves: it is more simply a form of self-promotion on external platforms, in a context that is strongly oriented towards the artistic product's commercialization.

The analysis that this research has dedicated to the ways in which artists present themselves, in order to be more exhaustive, had to carry out a parallel analysis of the platform governance models that determine the methods by which public-artist, user-platform, and artist-platform relationships unravel, in the light of the legitimation systems promoted by the platforms themselves.

Legitimation in the platform of Opensea appears to be heavily mediated by the platform's active engagement in the presentation of works: in the sections of "Notable Drops", the "Trending" filter and the daily selection of works of art, NFTs among which also creative works that classify as "Art" within the platform are selected according to undisclosed criteria and processes. On the other hand, HEN's less commercially oriented context appears to reshape legitimation frameworks limiting them to the boundaries of its platform and community. Through the "hDAO" default filter which sorts artworks according to the community's input (as they can upvote by donating "hDAO" tokens to artworks they appreciate or want to support), the platform of HEN does not only propose an alternative ranking and promoting system for artworks in its landing page, it also directly promotes alternative legitimation frameworks in which the artists' engagement on external social media for social status probably become less necessary and preceded by a search for social recognition within the platform itself.

Referring to the general statutes of pre-blockchain art, Crane (1976) suggests, as Bourdieu did, that since critics play a central role in the legitimation process, their "guardian role" is built and developed in parallel with incumbent entrepreneurs and dominant voices within a specific social context. All are, in the final analysis, co-responsible for the dissemination of art.

Applying Crane's (1976) thought to the dynamics that regulate the NFT market proposed by HEN and using his own words, in HEN the "dominant voices" and "incumbent entrepreneurs" are the collectors; the "function of guardians" they can perform is understandably stronger and at the same time more volatile because the legitimation frameworks can vary significantly due to their exposure to the fickle opinions of the public. On the other hand, while being explicitly oriented toward economic exchange and to the commercial dimension of art typical of the traditional art world, Opensea entrusts a ranking model which is inevitably more static: the results of sale as signifiers of appreciation of the work by the community. In most cases, such criteria also determine the legitimizing mechanisms of traditional art, as external sources (such as "Art Blocks Curated" on Opensea or very important galleries or collectors in contemporary art) have the power to impose an artist or a movement, making it difficult, if not impossible, to modify those legitimations.

Although Opensea affirms not making any claims about the legitimacy of artists, besides its navigational model, as demonstrated in this research, many aspects linked to its moderatory, regulatory and curatory practices appear to suggest the opposite.

A first substantial difference relies on the processes to confer users the verification badge. Opensea, unlike HEN, exclusively assigns it to artists who have exceeded a certain minimum sales threshold, established by Opensea (75 ETH), and have a significant presence on social media.

The verification badge that certifies the economic result and the social following are the two success criteria typical of the capitalist model, of which the platform of Opensea appears to be part of. Somehow, it is especially these two indicators that define and further support the legitimation model chosen by Opensea. It is a legitimation process that once again confirms the purely economic and market-oriented inclination of the platform, a legitimation that seems to adhere to the definition of "a state reached [...] when a source carefully assesses a new entrepreneur's propriety and then makes a judgment about the appropriateness of that entrepreneur in a social system. " (Kackovic & Wijnberg, 2020, p. 1093).
HEN, on the other side, withdraws from the role of mediator: the value of the verification badge is almost irrelevant, mainly representing a tool that helps users discern from "copyminters" or stolen identities. In the necessary steps required to verify one's users, HEN instead does not specify any criteria to meet, if not to verify their profiles on the platform by linking them to the corresponding Twitter profiles to verify their identity.

In addition to the unclear processes through which artworks in the platform of Opensea are selected and promoted across numerous sections, also the mechanisms through which Opensea can decide to freeze or disable accounts are not clearly stated. While HEN (that ultimately aimed at creating a decentralized voting system for making such decisions through on-chain trials) across its terms of service on the GitHub repository states clearly the criteria on which such decisions are taken, Opensea does not provide such detailed criteria stating it can decide at its discretion to delete accounts and therefore not accept or remove artists or works from its platform.
HEN is based on a governance model that invites the user to support many aspects of decisions and developments of the platform. Of course, this availability does not make HEN a democratically managed platform, because, even though the platform explicates, in its terms of service, the criteria through which such decisions are taken and promotes a vision of a community-built environment, the unaccomplished full decentralization through the complete developments of the Lima's visions for the "hDAO" token forced its founder to keep the final say on every decision.

Considering how in their different visions both platforms are ultimately centralized, one is brought to wonder what are the limits of the artist's freedom in a centrally managed crypto-digital environment.
However, overall the two platforms represent substantially different legitimation frameworks. The platform of HEN, being built on the Tezos blockchain, competed in the NFT space on a vision distant from an increasing commodification of art, conceived as a speculative financial tool (Price & Kuo, 2021). By proposing an alternative space for crypto artists and detaching from its role as a mediator, it created a space and promoted a vision for the development of the NFT phenomenon outside of the profit-driven logic that characterizes most of the popular corporate platforms in the NFT ecosystem. On the other hand, the legitimation frameworks that characterize the social context of Opensea highlight its role as an intermediary; as gatekeepers of artistic activity. As a platform of a corporate statute, from its navigational models (actively promoting artworks) to a

standardized taste in the aesthetics of artworks and artists' active engagement in social media to acquire social status, it does not appear to completely distance from the economic-oriented and profit-driven logics that characterize the traditional art market.

Although HEN's original platform after its shutdown has not been part of the NFT ecosystem anymore since November 2021 it has left a significant mark on the NFT community, especially in the Tezos ecosystem, sparking the hope for a possibility of further decentralization of the NFT ecosystem.
If on one side HEN appeared to engage in the promotion of a decentralized environment developing a system for the presentation of artworks based on a distributed voting system, it also appears clear how its instant rise in popularity from its launch is representative of the NFT enthusiasts' (and in particular artists engaged in the production of crypto art) desires and necessities to escape from the conventional mediated frameworks that govern the art world, whether traditional art galleries or digital platforms.
The later developments fueled by the community's contribution are an example of the impact that this platform (together with the alternative legitimation frameworks it suggested for artists in its vision for decentralization) has left in the field by opposing to traditional DeFi (decentralized finance) structures in Web 3.0, dominated by centralized corporate services that mediate interaction with the blockchain.
Being aware that legitimation frameworks in the digital and crypto art world are complex mechanisms that do not limit solely to the platform and online social spaces, this research suggested an interpretation of the legitimation systems that characterize the two platforms from a wide perspective: through the ways in which platforms mediate user's activity as well as through the social frameworks they promote and repurpose in social engagement across the platform of Twitter. However, acquiring a full understanding of the frameworks of legitimation that govern the two platforms, especially for an economic-oriented reality such as that of Opensea, would require going beyond the consideration of the digital spaces (platforms and social media activity) that govern the legitimation of artists in the NFT space. As legitimation of artists could be approached from a variety of perspectives due to the complexity of this social phenomenon, future research could be invested in considering also how the same NFT platforms are legitimized and how artists' legitimation occurs outside of the frameworks of these platforms, to what extent the art dealers and critics of the traditional art world are influencing and inserting themselves in the NFT space.
Although the research aimed to identify the social contexts in which legitimation frameworks of the two platforms are shaped, the approach to social aesthetics finds its limitations. The analysis of the aesthetics of artworks (of which authors have also been selected for the section of aesthetics of artists' presentation) focusing on the most acclaimed artists on both platforms has left aside many other less popular artists that, even if in weaker forms, engage in artistic activity and compete in these platforms at different levels of legitimacy. Identifying these different spheres in which legitimation occurs could unveil, especially in the economic-oriented reality of Opensea, structures of legitimation that although subordinated to the ones determining the success of most acclaimed artworks on the platform, still exist and are part of these ecosystems.

In the awareness of the vision introduced by the platform of HEN, in an ecosystem dominated by corporate centralized services, as appears to be the case in the NFT space, this thesis, by comparing two opposed realities and visions, sets the ground for future research to investigate the further developments in this ecosystem in light of autonomization of artists and artists' empowerment in the future developments of crypto art. Although the advancements of HEN's community, ultimately shifting to the Teia platform (latest and probably final landing of the community of the original platform of Hic et Nunc), continue to challenge centralized, proprietary, and opaque blockchain-based Web platforms, they inherit their success from the initial vision introduced by Lima. As this research focuses on the comparison of Opensea and the alternative platform introduced by Lima, analyzing the community in the fully decentralized platform of Teia, as a continuation of the visions introduced by HEN, could further shed light on the challenges and horizons for crypto artists' legitimation, in an ever-changing landscape.

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